

Enhancing reliability of integrated electricity-water systems with data centers via decentralized robust planning considering decision-dependent uncertainties

Chen Liang, Bo Zeng^{*}, Yueyi Lei, Zhu Wen, Jiayi Zhang

State Key Laboratory of Alternate Electrical Power System with Renewable Energy Sources (North China Electric Power University), Beijing 102206, China

HIGHLIGHTS

- Decentralized robust planning enhances reliability of coupled IDC-IEWS systems.
- Multi-agent coordination is enabled under decision-dependent and decision-independent uncertainties.
- A dual reliability enhancement model is designed using IDC backup power and water tank.
- Composite algorithm ensures tractable decentralized computation.
- Effectiveness and adaptability are verified through sensitivity analysis on key parameters.

ARTICLE INFO

Keywords:

Internet data center
Integrated electricity-water system
Robust optimization
Decision-dependent uncertainty
Coordination
Reliability enhancement

ABSTRACT

The rapid expansion of Internet Data Centers (IDCs) has significantly intensified the operational pressure on urban electricity and water infrastructures, necessitating the development of cross-system collaborative planning mechanisms to enhance overall system reliability. This paper proposes a tri-level robust planning framework for the coordinated optimization of IDCs and the integrated electricity-water system (IEWS) under multiple uncertainties. The upper-level model optimizes IDC resource deployment and IEWS infrastructure reinforcement investments. The middle-level model constructs decision-dependent uncertainty (DDU) and decision-independent uncertainty (DIU) sets to jointly capture the endogenous and exogenous coupling mechanisms among load fluctuations, power grid failure scenarios, and renewable energy outputs. The lower-level model addresses system operation scheduling under worst-case disturbances to ensure operational safety and cost-effectiveness. To enhance tractability and convergence efficiency, a hybrid solution approach is developed by integrating Parametric Column-and-Constraint Generation (PC&CG), Analytical Target Cascading (ATC), and Alternating Optimization Process (AOP), enabling synchronized decentralized solving across heterogeneous stakeholders. Case studies demonstrate that the proposed framework significantly improves system reliability under multiple disturbance scenarios, reducing electricity and water load shedding rates from 4.31% to 1.29% and from 6.33% to 2.15%, respectively, while enhancing system adaptability and robustness. This work provides a scalable and privacy-preserving theoretical foundation and methodological tool for robust multi-agent co-planning in integrated IDC-IEWS environments.

1. Introduction

As urban modernization accelerates, power and water supply systems serve as two critical lifeline infrastructures that play a pivotal role in ensuring the secure operation of future cities. The widespread adoption of electrically powered equipment in water networks has tightened

the coupling between the two systems, forming an interdependent Integrated Electricity-Water System (IEWS) whose security and reliability are of growing concern. Meanwhile, Internet Data Centers (IDCs), as large-scale consumers of both electricity and water, are projected to exceed 400 TWh in electricity use and $3 \times 10^9 \text{ m}^3$ in water use by 2030 [1]. The large-scale integration of IDCs not only increases the regulation pressure on the power system but also threatens the safety of urban

^{*} Corresponding author.

E-mail address: alosecity@126.com (B. Zeng).

<https://doi.org/10.1016/j.apenergy.2026.127574>

Received 13 November 2025; Received in revised form 2 February 2026; Accepted 17 February 2026

Available online 2 March 2026

0306-2619/© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Nomenclature

Abbreviations

IEWS	Integrated Electricity-Water System
IDC	Internet Data Center
RL/DL	Real-time/Delay-tolerant loads
SDRP	Synchronously Decentralized Robust Planning
ATC	Analytical target cascading
RES	Renewable energy sources
DDU	Decision-dependent uncertainty
DIU	Decision-independent uncertainty
KKT	Karush-Kuhn-Tucker
WT	Wind turbine

Indices and sets

$t \in \Omega^T$	Time periods
$y \in \Omega^Y$	Planning stages
$d \in \Omega^D$	Internet Data Center node
$(i, j) \in \Omega^L$	Power line index
$g \in \Omega^G$	Generator index
$i \in \Omega^I$	Bus index
$p \in \Omega^P$	Pipeline index
$w \in \Omega^W$	Water node index

Parameters

Δt	Duration of each time interval [h]
M	Large positive constant for Big-M method
ζ	Discount rate for investment planning
$c_{d,y,t}^{ele}/c_{d,y,t}^{wat}$	IDC and IEWS electricity/water transaction prices [¥/kWh, ¥/m ³]
γ/ρ	Lagrangian penalty multipliers for ATC algorithm
$p_{DC,max}^{DC,max}$	IDC and IEWS electricity transmission limits [kW]
$f_{DC,max}^{DC,max}/f_{DC,max}^{RLT}$	IDC and IEWS water transmission limits [m ³ /s]
$\bar{D}_{d,y,t}^0$	Initial data load [PB]
$D_{d,y,t}^{DLO}/D_{d,y,t}^{RLO}$	Initial DL/RL [PB]
χ^{DL}/χ^{RL}	Proportions of delay-tolerant / real-time loads
ψ	Processing rate [PB/h]
T^{Delay}	Maximum allowable delay [s]
ϕ	Server configuration quantity [kW]
p^{ser-p}/p^{ser-i}	Server peak/silent power [kW]
$T_{d,y,t}^{out}$	IDC outdoor temperature [°C]
R_d	IDC thermal resistance [°C/kW]
η^{ser}	Cooling coefficient
μ_1/μ_2	Cooling load coefficient/equipment activation coefficient
T_{in}/T_{in}^{in}	IDC indoor minimum/maximum temperature [°C]
η^{lo}	Evaporation coefficient [m ³ /kWh]
$\bar{h}^{po}/\bar{h}^{po}$	IDC tank minimum/maximum head [m]
η^{ch}/η^{ch}	Charging/discharging coefficient [¥/unit]
c^{ser}/c^{co}	Unit investment cost of servers and chillers [¥/unit]
c^{DR}	Unit cost for shifting data loads [¥/PB]
c_{ij}^{al}/c_g^{ag}	Unit reinforcement cost of lines/DGs [¥/unit]
$c_{y,t}^{grid}/c_{y,t}^{hw}$	Electricity/water purchase price from the main grid/water source [¥/kWh, ¥/m ³]
$c^{e,curt}/c^{w,curt}$	Penalty price for power/water load curtailment

$[\text{¥/kWh}, \text{¥/m}^3]$	
$c^{d,curt}$	Penalty cost for data load shedding [¥/PB]
ζ	Discount rate for investment planning
p_{ij}^{al}/Q_{ij}^{al}	Active/reactive power capacity expansion for line reinforcement [kW, kvar]
$\underline{V}/\bar{V}$	Nodal voltage lower/upper limit [kV]
$\underline{p}_g^{ag}/\bar{p}_g^{ag}$	Generator reinforcement capacity expansion [kW]
$\underline{h}_p^{\Delta,ini}/\bar{h}_p^{\Delta,ini}$	Inlet head lower/upper limit of the water pipeline [m]
$\underline{h}_p^{\Delta,ter}/\bar{h}_p^{\Delta,ter}$	Outlet head lower/upper limit of the water pipeline [m]
a_p^{lwp}/b_p^{lwp}	Quadratic/linear coefficient of pump power to flow ratio
R_p^{lwp}	Hydraulic resistance [s ² /m ⁵]
$\Gamma_{i,y}^{res}/\Gamma_{i,y}^{el}/\Gamma_{w,y}^{wl}$	Outputs of RES/electricity load/water load upper limit
$\Gamma_y^{al}/\Gamma_y^{ag}$	Maximum allowable number of transmission line/generator faults
Variables	
$A_{d,y}^{ser}/A_{d,y}^{co}$	Number of added servers/chillers
$x_{d,y}^{ser}/x_{d,y}^{co}$	Number of servers/chillers
$p_{d,y,t}^{DC}/f_{d,y,t}^{DC}$	Electricity/water transaction volume between IDC and IEWS
$D_{d,y,t}^{DLT}/D_{d,y,t}^{RLT}$	Temporal/spatial shifting volume of data load
$D_{d,y,t}$	Amount of data load served
$D_{d,y,t}^{curt}$	Amount of data load curtailed
$n_{d,y,t}^{ser-on}$	Number of active servers
$p_{d,y,t}^{ser}/p_{d,y,t}^{co}$	Server/chiller power
$p_{d,y,t}^{ch}/p_{d,y,t}^{dch}$	Charging/discharging power of energy storage
$H_{d,y,t}^{co}$	Cooling energy
$f_{d,y,t}^{lo}$	Water evaporation volume
$T_{d,y,t}^{in}$	IDC indoor temperature
$h_{d,y,t}^{po}$	Water head of the IDC tank
$E_{d,y,t}$	SoC of energy storage
$A_{l,y}^{al}/A_{g,y}^{ag}$	Reinforcement status of the line/generator in year y
$x_{l,y}^{al}/x_{g,y}^{ag}$	Status of the line/generator
$p_{y,t}^{grid}/f_{y,t}^{hw}$	Electricity/water purchased from the main grid/water source
$p_{i,y,t}^{curt}/f_{w,y,t}^{curt}$	Electricity/water load curtailment volume
$p_{i,y,t}^{res}/p_{i,y,t}^{gen}$	RES/generator power output
$p_{i,y,t}^{lwp}$	Power consumption of water pumps
$p_{i,y,t}/Q_{i,y,t}$	Active/reactive power of the line
$Q_{i,y,t}^{res}/Q_{i,y,t}^{gen}$	Reactive power output of RES/generators
$V_{i,y,t}$	Bus voltage
$\bar{u}_{ij,y,t}/\bar{v}_{g,y,t}$	Line/generator failure condition
$\bar{P}_{i,y,t}^{res}$	Maximum RES power output
$h_{p,y,t}^{\Delta,ini}/h_{p,y,t}^{\Delta,ter}$	Inlet/outlet head of the water pipeline
$h_{p,y,t}^{\Delta,hl}$	Water head loss in the pipe
$f_{p,y,t}^{lwp}$	Pipeline water flow
$\bar{f}_{w,y,t}^L$	Water load
$u_{i,y,t}^{res,+}/u_{i,y,t}^{res,-}/u_{i,y,t}^{el,+}/u_{i,y,t}^{el,-}/u_{w,y,t}^{wl,+}/u_{w,y,t}^{wl,-}$	Uncertainty fluctuation variable

water supply. Within the tri-coupled context of information, power, and water flows, coordinating the integration and operation of IDCs and the IEWS to ensure secure supply and enhance the reliability of the overall system has become a significant challenge.

To coordinate the integration of large-scale IDCs into power systems,

increasing studies have focused on exploiting the flexibility of IDCs' electrical load demands, particularly the modeling of various data workloads and internal equipment characteristics to enhance their operational flexibility. Earlier works treated IDCs as controllable loads and utilized for demand response program; Ref. [2] modeled them as

responsive virtual power plants, enabling peak shaving through real-time adjustment of computing loads. Subsequent researches, such as Ref. [3], incorporated the time-varying characteristics of cooling loads and developed integrated scheduling models that account for both computing and cooling demands, improving dispatch accuracy and operational adaptability. Ref. [4] examined the capability of IDCs to perform rapid load shifting under extreme operating conditions, proposing flexible scheduling strategies that enhance grid stability in contingency scenarios. In addition, Refs. [5, 6] suggested leveraging IDCs' backup generation and emergency power units as supplementary resources in contingency scenarios, where distributed load regulation and fast-start capabilities enable IDCs to maintain local operations and, when necessary, supply power back to the grid to support peak shaving, self-supply, and fault recovery.

In contrast to the abovementioned works focusing on electrical flexibility, only a limited number of studies have discussed the water-side flexibility of IDCs. Ref. [7] investigated how water resource availability affects IDC siting decisions, emphasizing the spatial correlation between water access and cooling efficiency. Ref. [8] modeled IDC cooling load as a dispatchable variable, enabling joint optimization with electrical loads and expanding the role of IDCs in system scheduling. Ref. [9] introduced chilled water storage within IDCs as an adjustable resource, proposing scheduling strategies for charging and discharging cooling tanks to balance internal and external cooling needs. Ref. [10] incorporated IDC cooling ponds into multi-period optimization, allowing water-side resource allocation to more flexibly respond to varying operational conditions and external system requirements.

To realize the coordinated operation between IDCs and IEWS, several studies have developed joint planning frameworks that exploit the flexibility potential of IDCs. Ref. [11] first incorporated IDCs into the unified planning of electric-water systems, optimizing overall performance by coordinating power allocation and water dispatch. Ref. [12] extended this work by introducing multi-energy coordination mechanisms, enabling the joint allocation of IDC's electrical and water demands with renewable generation and energy storage resources. Ref. [13] further expanded the co-planning approach to integrated energy systems, emphasizing energy complementarity and cross-system optimization to improve resource utilization. Ref. [14] developed the Multi-Objective Scheduling and Interactive Control (MOSAIC) framework, a multi-objective optimization model for sustainable datacenter management that simultaneously coordinates electricity use, cooling demand, and water consumption across geographically distributed facilities. Ref. [15] integrated reliability metrics into the co-planning objectives, strategically allocating redundant resources and backup capacities both inside and outside the IDC to significantly strengthen the system's ability to sustain energy and water supply under faults and extreme events.

For the practical implementation of IDC-IEWS collaborative planning, effective management of multi-source uncertainties is critical to ensure the validity of planning solutions. Specifically, IDC loads exhibit pronounced volatility due to variations in task scheduling, user access patterns, and cooling mode transitions, often manifesting as short-duration, high-magnitude disturbances [16]. Ref. [17] employed a multi-state Markov chain to model the coupling between computational and cooling loads, enabling state-based load uncertainty characterization. Based on this, Ref. [18] incorporated the nonlinear responses of cooling loads to environmental conditions and IT equipment utilization, enhancing thermal disturbance prediction via a generalized regression neural network. On the power grid side, structural uncertainties—such as line outages, node failures, and cascading failures—have been addressed using nodal-importance-based failure probability distribution models [19], hybrid fuzzy-interval-probabilistic approaches with hierarchical fault path screening [20], and multi-stage fault evolution models that integrate scenario trees with Monte Carlo simulations [21] to strengthen dispatch strategies under extreme disturbance scenarios. Ref. [22] introduced a distributed robust optimization framework using

Alternating Direction Method of Multipliers (ADMM) to address uncertainties in multi-agent systems, improving the system's ability to handle large-scale problems and accelerating convergence in decision-making processes.

Although many studies have considered data load fluctuations and grid disturbances in scheduling models, most of them treated uncertainties as Decision-Independent Uncertainty (DIU) in their research works, neglecting the potential coupling effect between uncertainties and system decisions. In practice, uncertainties are often shaped by scheduling schemes and resource allocation, exhibiting Decision-Dependent Uncertainty (DDU) [23]. Ref. [24] first introduced DDU in a stochastic optimization framework, establishing a functional mapping between decision variables and uncertainty sets. Ref. [25] extended this to a multi-stage robust optimization model with state-feedback logic, enabling dynamic adaptation to evolving uncertainties. In the energy sector, Ref. [26] applied DDU to a combined heat and power system, reconstructing uncertainty sets based on operating states and disturbance intensities. Ref. [27] proposed a hierarchical scenario-based framework to integrate dynamic load feedback for enhanced dispatch responsiveness. Since DDU often yields complex uncertainty sets that challenge computational efficiency, Ref. [28] developed a Parametric Column-and-Constraint Generation (PC&CG) approach, decomposing the dynamic uncertainty space into a finite adjustable set, thereby balancing modeling fidelity with algorithmic scalability.

Despite growing research efforts on Internet Data Center and Integrated Electricity-Water System (IDC-IEWS) coupled systems, several critical research **gaps** remain in the domains of coordinated planning and reliability enhancement, making it difficult to fully achieve cross-system collaboration in real-world applications.

First, building upon the above studies that primarily focus on IDC flexibility on the electricity side, there remains a clear limitation when examining its potential roles in other public utility networks. Most existing research on IDC's contribution to reliability enhancement or emergency dispatch has centered on its support for the power system, such as peak shaving with backup power or maintaining self-supply under fault conditions [5,6]. Such models typically treat the IDC as a responsive electrical load or flexible energy node, with its support capacity confined to the electrical domain. **In contrast, the capability of IDCs to provide emergency water back-feed to municipal networks remains largely unexplored. Specifically, using internal storage to compensate for pump station outages lacks systematic modeling.** In reality, many critical water supply facilities (e.g., secondary booster pumps, reservoir inlet/outlet pumps) are highly dependent on electricity, and large-scale grid failures without external power support can trigger secondary water supply interruptions, threatening the stability of public services. Existing planning models often treat IDCs as purely power-side adjustment units, without capturing their reverse water-supply interactions with the water system, thereby limiting system recovery capability and underutilizing the value of cross-domain resource coordination.

Second, even within the limited number of studies that consider IDC-IEWS cross-domain coordination, most still adopt a centralized planning framework, assuming that IDC and IEWS are jointly managed and invested in by a single entity. While this assumption simplifies modeling, it deviates significantly from real-world conditions, where IEWS and IDC are often operated by independent entities with heterogeneous objectives and distinct security boundaries [7,11]. Data center operators (DCOs) are typically more focused on cost and energy efficiency, while the power and water supply companies prioritize the stability and quality of electricity and water supply. **In such a multi-stakeholder environment, the assumption of a single decision-maker performing joint optimization becomes unrealistic.** Although recent integrated energy system research involving IDCs has introduced game theory and distributed optimization tools to address multi-party coordination [29–31], such modeling has largely been limited to the operational stage. There is still a lack of planning-stage mechanisms for

multi-party reliability coordination that account for diverse stakeholder needs, leading to service coordination, resource allocation, and reliability assurance being reactive responses rather than proactively designed features to enhance system robustness.

Third, under the multi-stakeholder, multi-energy coordination context, the representation of uncertainties in existing models remains insufficient. While some recent studies have introduced DDU modeling approaches, enabling the uncertainty set to evolve with scheduling strategies, most work remains focused on single-energy domains, such as renewable generation output or price fluctuations [28,32,33]. **There is still a lack of DDU representation for data load disturbances and distribution network fault scenarios. The structures of both vary dynamically with planning decisions.** In electricity-water coupled systems, IDC electricity consumption patterns and network reinforcement decisions directly influence load profiles and vulnerability distributions. Treating these uncertainties as static, exogenous inputs fails to capture the feedback-control mechanism between system structure and uncertainty space, which may result in overly conservative planning or unbalanced resource allocation. A summary of previous studies and this paper is provided in Table 1.

To address the challenges identified above, this paper proposes a synchronized decentralized robust optimization planning framework for IDC-IEWS coupled system, aimed at achieving coordinated decision-making among multiple stakeholders under reliability objectives. The framework adopts a tri-level decomposition strategy based on the Min-Max-Min robust optimization formulation. This structure is explicitly designed to address the computational intractability of identifying the worst-case realization within the complex DDU sets. Accordingly, the problem is modeled as a three-level nested Min-Max-Min robust optimization model: (1) the upper-level planning layer determines the investment strategies for IDC emergency capacity expansion, IEWS equipment reinforcement, and bi-directional electricity-water supply interfaces; (2) the middle-level uncertainty modeling layer selects the most adverse load demand, renewable output, and fault disturbance scenarios from the given DDU and DIU sets; (3) the lower-level scheduling layer executes the joint secure scheduling of the system under extreme disturbances to ensure critical load supply and constraint satisfaction. To efficiently solve this multi-layer coupled robust model, this paper designs a composite optimization strategy: the inner-layer robustness is driven by the PC&CG algorithm, the multi-agent coupling part is handled by the Analytical Target Cascading (ATC) mechanism for boundary adjustment and information protection between IDC and IEWS, and the upper-level investment integer variables are iteratively optimized via the Alternating Optimization Process (AOP) to converge toward a solution. The overall framework approximates the global optimal solution while preserving data privacy for each stakeholder. The main innovations and contributions of this paper are reflected in the following three aspects:

(1) **A unified IDC dual-support model based on emergency power supply and water tank capacity is developed.** This model jointly characterizes multi-dimensional operational flexibilities, such as temporal load shifting, spatial demand regulation, and cross-domain power-water resource switching. The model systematically coordinates

IDC reverse power feeding and reverse water supply to provide multi-path support under distribution network failure scenarios, ensuring the continuous operation of data centers while maintaining the stability of critical grid nodes and water infrastructure.

(2) **A decentralized robust planning framework is proposed for the reliability enhancement of IDC-IEWS coupled system**, overcoming the drawbacks of traditional isolated planning paradigms, such as information silos and lack of interactive negotiation.

(3) **An uncertainty characterization considering both DDU and DIU is introduced.** A novel joint planning scenario with DDU modeling is developed, in which IDC investment decisions (server deployment) affect the bounds of data load fluctuations, and IEWS reinforcement decisions (line and generator upgrades) influence fault scenarios. This linkage enables the uncertainty set to adapt dynamically with planning strategies, thereby reducing conservatism while enhancing flexibility and robustness.

The structure of this paper is organized as follows: Section 2 introduces the coupling structure and modeling of IDC and IEWS. Section 3 presents the synchronized decentralized three-level robust optimization framework and defines the decision-dependent uncertainty sets. Section 4 designs a distributed solution algorithm based on PC&CG, ATC, and AOP. Section 5 validates the model through case studies, comparing system reliability and scheduling economy under different strategies. Finally, Section 6 summarizes the paper and discusses potential directions for future research.

2. System architecture and model formulation

2.1. System components and coupling mechanism

The system investigated in this study comprises two primary components: the Integrated Electricity-Water System (IEWS) of the city and the Internet Data Center (IDC). The IEWS consists of a distribution power network and a water distribution network. The power network transmits electricity from the upstream grid and distributed renewable energy sources, such as wind power and photovoltaics, to various urban electricity consumers and the IDC. Meanwhile, the water network delivers water resources from sources such as water treatment plants and reservoirs to urban water users and the IDC via pumps, pipelines, and storage tanks. As critical equipment within the water network, pumps ensure adequate water pressure and flow rates while consuming significant amounts of electrical energy, thereby physically coupling electricity and water resources. As a supply-side infrastructure in urban areas, the IEWS provides both electricity and water resources to the IDC. It dynamically adjusts its power transmission and water supply capacities in response to fluctuating demands from urban loads and the IDC, thereby enabling the coordinated supply of energy and water resources.

The IDC, as a data processing infrastructure characterized by intensive energy and water consumption, comprises three key subsystems: server clusters, cooling systems, and energy storage units. Server clusters continuously consume electricity to support ongoing computational loads, while the cooling system dissipates heat through chillers, cooling towers, water pumps, and terminal air-conditioning units. Cooling

Table 1

A summary of previous studies and the present work.

Reference	Consideration of Faults	Research field	Model	IDC Support		Uncertainty	DIU	DDU
				Power	Water			
[1–6]	✗	Operation	Centralized	✗	✗	✗	✗	✗
[7–10]	✗	Operation	Centralized	✗	✗	✗	✗	✗
[11–15]	✓	Planning	Centralized	✓	✗	✗	✗	✗
[16–19]	✓	Operation	Centralized	✗	✗	✗	✗	✗
[20–22]	✗	Operation	Centralized	✗	✗	✓	✓	✗
[23–28]	✓	Planning	Centralized	✗	✗	✓	✗	✓
[29–31]	✗	Operation	Decentralized	✗	✗	✗	✗	✗
Proposed	✓	Planning	Decentralized	✓	✓	✓	✓	✓

towers, in particular, evaporate large amounts of water, and water pumps maintain the circulation of cooling water, physically linking water consumption with power usage. Battery storage systems are deployed for electrical energy storage and release, while water storage tanks regulate water resource supply, thereby enhancing the IDC's operational flexibility regarding electricity and water utilization. The IDC obtains electricity and water via the power distribution network and water distribution network, respectively, and dynamically adjusts server operations, cooling demands, and storage strategies according to its load status. These adjustments, in turn, influence the supply levels of the IEWS, establishing a bidirectional physical coupling characterized by interactive “electricity-water” flows. Furthermore, on the information flow level, data users submit service requests to the IDC through the internet, generating both real-time loads (RL), which require immediate processing, and delay-tolerant loads (DL), which can be flexibly scheduled in time and space to achieve load balancing and resource optimization. Guided by Service Level Agreements (SLA), DCO dynamically manages RL and DL scheduling strategies by considering server status, cooling capacity, and resource availability, ensuring service quality while continuously adjusting the operation of servers and cooling systems. The information flow thus directly determines the IDC's demand for electricity and water, forming a real-time interactive supply-demand relationship with the IEWS and giving rise to a physically coupled architecture across three interconnected domains: information flow, electricity flow, and water flow, as shown in Fig. 1.

It is important to note that the IDCs considered in this study are primarily connected to distribution networks and represent the growing class of Edge Data Centers. While their individual capacity is relatively smaller than hyperscale facilities connected to transmission grids, their capability to support the network is justified by their strategic location and localized backup redundancy. Specifically, these IDCs strictly adhere to high reliability standards such as Tier III or IV and thus possess substantial onsite backup resources. This redundancy allows them to

provide targeted emergency support to local critical infrastructures, such as municipal water pumps, at the distribution transformer or district level. By preventing the failure of these key nodes during faults, they effectively block the propagation of cascading failures, thereby enhancing the localized resilience of the IEWS and indirectly contributing to the stability of the wider urban infrastructure.

2.2. Power-water failures and IDC response

As illustrated in Fig. 1, water pumps within the urban IEWS require continuous power from the distribution network to maintain normal pressurization and water delivery. When portions of the power network experience outages due to equipment faults or external disturbances, the associated water pumps lose power and cease operation. This leads to an immediate drop in hydraulic head across the water network. Given the serial connectivity of the water distribution pipelines, upstream pressure losses propagate downstream, affecting supply to terminal nodes. End-users located at distal pipeline ends or in elevated regions are particularly vulnerable to insufficient water pressure, as they rely on sustained pressurization from upstream segments.

In addition to securing its own water and electricity supply, the IDC enhances urban network reliability by actively supporting the broader distribution systems. Specifically, the IDC is equipped with pumping facilities that enable it to discharge water from its internal storage tanks back into critical nodes of the municipal water distribution network. This mechanism compensates for localized water pressure deficits and supply shortages caused by pump outages due to power failures. Similar reverse water supply mechanisms have been successfully applied in other water resource management systems, such as the use of water storage systems for emergency supply during power outages, as seen in Poland's approach [34]. Simultaneously, the IDC's battery energy storage systems can discharge power back into the distribution grid during electrical faults, providing supportive capacity to local power nodes,

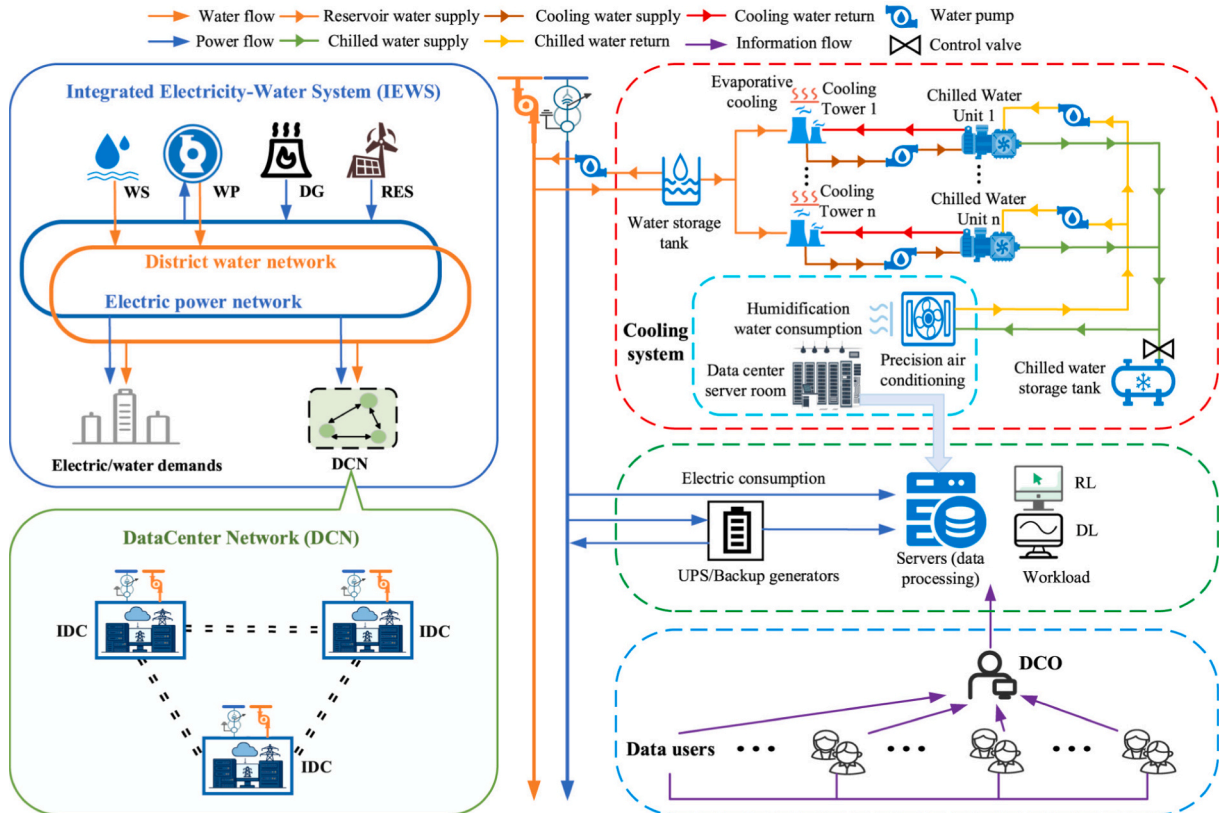


Fig. 1. Structures of the IEWS and IDCs.

alleviating supply gaps, and reducing stress on the distribution network. Through two coordinated mechanisms, namely water pumping for external supply and battery power backfeed, the IDC offers bidirectional support to the IEWS on both the electrical and hydraulic sides. This dual-domain assistance mitigates the risk of cross-sector cascading failures triggered by power outages, thereby significantly enhancing the overall reliability of the IEWS. Technically, this bidirectional support aligns with emerging industry practices, such as IDCs operating as Virtual Power Plants or participating in eco-industrial circular water management. From an engineering planning perspective, the proposed framework serves as a quantitative tool to evaluate the economic trade-offs between investing in such interactive coupling interfaces versus the traditional alternative of expanding utility infrastructure redundancy, such as increasing backup power and water supply capacities.

Remark 1. While the assumption of reverse water supply from IDCs is hypothetical, it can be technically feasible in scenarios where dedicated reclaimed-water pipelines or intermediate storage tanks are deployed near industrial parks. In practice, such integration would require bidirectional pumping units, backflow prevention devices, and water-quality control facilities to ensure safe operation. Therefore, in this macroscopic planning model, we strictly focus on the hydraulic feasibility (constrained by network topology and pressure limits), treating water quality compliance as a prerequisite satisfied by these engineering facilities. This assumption thus represents a forward-looking scenario to assess the potential of IDC–municipal water interaction under future infrastructure upgrades.

3. Robust collaborative planning model formulation

The DDU-based Synchronously Decentralized Robust Planning (SDRP) model of the IEWS and IDC is illustrated in Fig. 2. Both the IEWS and IDC are treated as independent stakeholders, each operating under its own set of uncertainties and making planning and design decisions autonomously. The objective function for each entity is to minimize its total cost, which comprises three components: investment cost, interaction cost, and operational cost.

$$F_{DC/IEWS} = \min \left(F_{DC/IEWS}^I + F_{DC/IEWS}^{II} + \max_{\min} F_{DC/IEWS}^{III} \right) \quad (1)$$

3.1. Three-level model structure and objective function of IDC

The first component of the IDC's objective function includes the capital investment costs associated with capacity expansions of servers and chillers. The second component accounts for interaction costs, which reflect electricity transactions with the power distribution network and water transactions with the water distribution network. The third component captures the annual operational costs under data load uncertainty. The formulation of the first component of the objective function, along with its associated constraints, is presented as follows:

$$F_{DC}^I = \sum_{y \in \Omega^Y} \frac{1}{(1 + \zeta)^{y-1}} \sum_{d \in \Omega^D} \left(c^{\text{ser}} A_{dy}^{\text{ser}} + c^{\text{co}} A_{dy}^{\text{co}} \right) \quad (2)$$

$$A_{dy+1}^A = x_{dy+1}^A - x_{dy}^A, \forall d \in \Omega^D, \forall y \in \Omega^Y, A \in \{\text{ser}, \text{co}\} \quad (3)$$

$$0 \leq A_{dy}^x, x_{dy}^A \leq x_d^{A, \max}, \forall d \in \Omega^D, \forall y \in \Omega^Y, A \in \{\text{ser}, \text{co}\} \quad (4)$$

Eq. (2) represents the annualized investment cost of the IDC, comprising the capital costs for newly installed servers and chillers. Eq. (3) describes the relationship between the available capacity of IDC equipment and the annual installation capacity. Eq. (4) imposes constraints on the maximum allowable installation capacity of each type of equipment.

The second component of the objective function, along with its associated constraints, is formulated as follows:

$$F_{DC}^{II} = \sum_{y \in \Omega^Y} \frac{N}{(1 + \zeta)^{y-1}} \sum_{t \in \Omega^T} \sum_{d \in \Omega^D} \left(C_{dy,t}^{\text{ele}} + C_{dy,t}^{\text{wat}} \right) \quad (5)$$

$$C_{dy,t}^{\text{ele}} = c_{dy,t}^{\text{ele}} P_{dy,t}^{\text{DC}} \Delta t, \forall d \in \Omega^D, \forall y \in \Omega^Y, \forall t \in \Omega^T \quad (6)$$

$$C_{dy,t}^{\text{wat}} = c_{dy,t}^{\text{wat}} f_{dy,t}^{\text{DC}} \Delta t, \forall d \in \Omega^D, \forall y \in \Omega^Y, \forall t \in \Omega^T \quad (7)$$

$$|P_{dy,t}^{\text{DC}}| \leq P_{dy,t}^{\text{DC}, \max}, -f_{dy,t}^{\text{DC}, \max} \leq f_{dy,t}^{\text{DC}} \leq f_{dy,t}^{\text{DC}, \max}, \forall d \in \Omega^D, \forall y \in \Omega^Y, \forall t \in \Omega^T \quad (8)$$

Eq. (5) defines the interaction cost, accounting for the expenses of electricity and water purchasing from the distribution networks. Eq. (6) captures the basic interaction cost associated with the power distribution network, while Eq. (7) reflects the interaction cost with the water distribution network. Eq. (8) imposes upper bounds on the power and

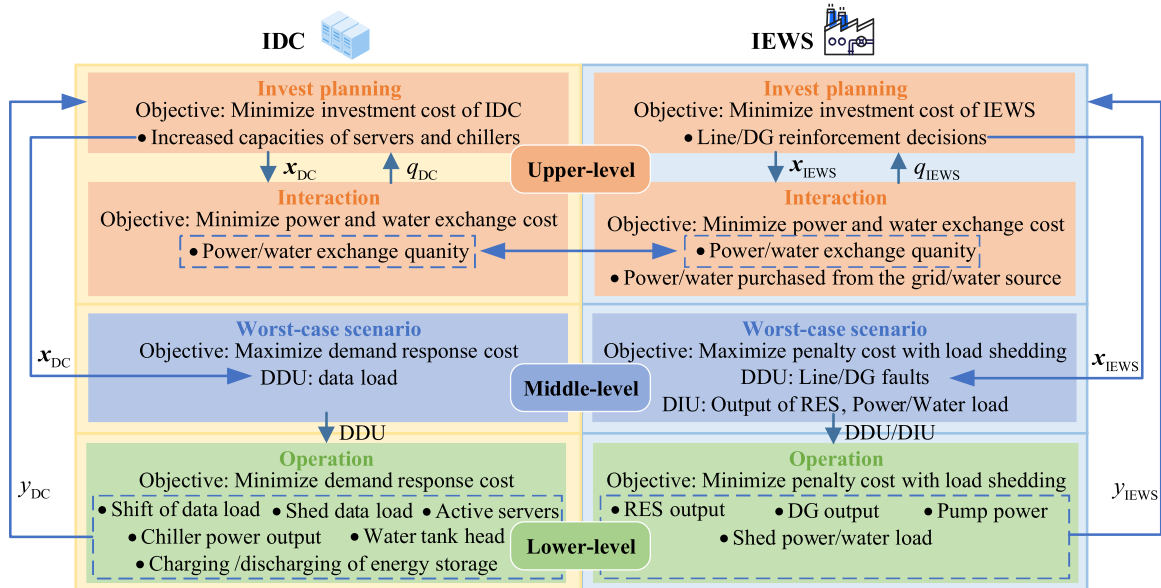


Fig. 2. SDRP considering DDU model.

water mass flow rates involved in interactions, ensuring that they do not exceed the maximum allowable interaction limits.

Interactive electricity and water flows are modeled as first-stage decisions to decouple multi-agent coordination from lower-level robust operations. As noted in [35], determining interaction variables in the second stage would necessitate negotiating exchanges for every uncertainty realization, leading to excessive computational burden and privacy violations. Therefore, establishing a robust interaction boundary in the planning stage is essential for the solvability and scalability of the decentralized framework.

The third component of the objective function, together with its associated constraints, is formulated as follows:

$$F_{DC}^{III} = \sum_{y \in \Omega^Y} \frac{N}{(1 + \zeta)^{y-1}} \sum_{t \in \Omega^T} \sum_{d \in \Omega^D} \left(\sum_{r \in \Omega^T} c^{DR} D_{d,y,t}^{DLT} (t - r) + c^{d,curt} D_{d,y,t}^{curt} \right) \Delta t \quad (9)$$

Eq. (9) formulates the operational recourse cost, specifically including the costs for data load shifting and penalty costs for shedding. Subject to the following operational constraints:

(1) Workload constraints. The constraints regarding the allocation of delay tolerant and real time data loads are formulated as follows:

$$D_{d,y,t}^{DLO} = \chi^{DL} \bar{D}_{d,y,t}^0, \forall d \in \Omega^D, \forall y \in \Omega^Y, \forall t \in \Omega^T \quad (10)$$

$$D_{d,y,t}^{DL} = D_{d,y,t}^{DLO} + \sum_{t'=1}^{t-1} D_{d,y,t'}^{DLT} - \sum_{t'=t+1}^T D_{d,y,t'}^{DLT}, \forall d \in \Omega^D, \forall y \in \Omega^Y, \forall t \in \Omega^T \quad (11)$$

$$0 \leq D_{d,y,t'}^{DLT} \leq D_{d,y,t}^{DLO}, \forall d \in \Omega^D, \forall y \in \Omega^Y, \forall t, t' \in \Omega^T \quad (12)$$

$$\sum_{t'=t+1}^T D_{d,y,t'}^{DLT} \leq D_{d,y,t}^{DLO}, \forall d \in \Omega^D, \forall y \in \Omega^Y, \forall t, t' \in \Omega^T \quad (13)$$

$$D_{d,y,t}^{RLO} = \chi^{RL} \bar{D}_{d,y,t}^0, \forall d \in \Omega^D, \forall y \in \Omega^Y, \forall t \in \Omega^T \quad (14)$$

$$D_{d,y,t}^{RL} = D_{d,y,t}^{RLO} - \sum_{d' \neq d, d' \in \Omega^D} D_{d',y,t}^{RLT}, \forall d \in \Omega^D, \forall y \in \Omega^Y, \forall t \in \Omega^T \quad (15)$$

$$0 \leq D_{d,y,t'}^{RLT}, \forall d \in \Omega^D, \forall y \in \Omega^Y, \forall t \in \Omega^T \quad (16)$$

$$0 \leq \sum_{d' \neq d, d' \in \Omega^D} \max(0, D_{d',y,t}^{RLT}) \leq D_{d,y,t}^{RLO}, \forall d \in \Omega^D, \forall y \in \Omega^Y, \forall t \in \Omega^T \quad (17)$$

$$D_{d,y,t} + D_{d,y,t}^{curt} = D_{d,y,t}^{DL} + D_{d,y,t}^{RL}, D_{d,y,t} \geq 0, \forall d \in \Omega^D, \forall y \in \Omega^Y, \forall t \in \Omega^T \quad (18)$$

$$1 / \left(\psi - \frac{D_{d,y,t}}{n_{d,y,t}^{ser-on}} \right) + \frac{1}{\psi} \leq T^{Delay}, \forall d \in \Omega^D, \forall y \in \Omega^Y, \forall t \in \Omega^T \quad (19)$$

$$0 \leq n_{d,y,t}^{ser-on} \leq (1 - \phi) x_{d,y,t}^{ser}, \forall d \in \Omega^D, \forall y \in \Omega^Y, \forall t \in \Omega^T \quad (20)$$

Eqs. (10)–(13) respectively describe the proportional allocation of delay-tolerant loads (DL), load balance constraints before and after response, temporal shifting characteristics, and total load migration limits. Eqs. (14)–(17) address the proportional allocation of real-time loads (RL), load migration balance constraints, spatial shifting characteristics, and constraints on the total amount of migratable load. Eq. (18) calculates the total information load that each IDC must process. Eq. (19) imposes limits on queueing and processing delays, ensuring they do not exceed the maximum allowable latency thresholds for loads. Eq. (20) constrains the number of servers activated.

(2) Server and cooling constraints. The physical operational limits coupling computational workload with power consumption and temperature are defined as:

$$P_{d,y,t}^{DC} = P_{d,y,t}^{ser} + P_{d,y,t}^{co} + P_{d,y,t}^{dch} - P_{d,y,t}^{ch}, \forall d \in \Omega^D, \forall y \in \Omega^Y, \forall t \in \Omega^T \quad (21)$$

$$P_{d,y,t}^{ser} = n_{d,y,t}^{ser-on} P_{d,y,t}^{ser-i} + (P_{d,y,t}^{ser-p} - P_{d,y,t}^{ser-i}) D_{d,y,t} / \psi, \forall d \in \Omega^D, \forall y \in \Omega^Y, \forall t \in \Omega^T \quad (22)$$

$$H_{d,y,t}^{co} = \eta^{ser} P_{d,y,t}^{ser} - \frac{T_{d,y,t}^{in} - T_{d,y,t}^{out}}{R_d}, \forall d \in \Omega^D, \forall y \in \Omega^Y, \forall t \in \Omega^T \quad (23)$$

$$P_{d,y,t}^{co} = \mu_1 H_{d,y,t}^{co} + \mu_2, P_{d,y,t}^{co} \leq x_{d,y,t}^{co}, \forall d \in \Omega^D, \forall y \in \Omega^Y, \forall t \in \Omega^T \quad (24)$$

$$\underline{T}^{in} \leq T_{d,y,t}^{in} \leq \bar{T}^{in}, \forall d \in \Omega^D, \forall y \in \Omega^Y, \forall t \in \Omega^T \quad (25)$$

$$f_{d,y,t}^{lo} = \eta^{lo} H_{d,y,t}^{co}, \forall d \in \Omega^D, \forall y \in \Omega^Y, \forall t \in \Omega^T \quad (26)$$

Eq. (21) represents the power balance within the IDC. Eq. (22) indicates that server power consumption is positively correlated with processing load and the number of active servers. Eq. (23) determines the required cooling capacity based on ambient temperature and server heat dissipation, while Eq. (24) expresses the proportional relationship between cooling power and cooling load, subject to chiller capacity limitations. Eq. (25) imposes temperature constraints within the IDC. Eq. (26) relates evaporative water consumption to the cooling load.

(3) Battery and tank constraints. The operational constraints governing the backup battery systems and water circulation tanks are presented below:

$$h_{d,y,t}^{po} = h_{d,y,t-1}^{po} + (f_{d,y,t}^{DC} - f_{d,y,t}^{lo}) / A_d^m, \forall d \in \Omega^D, \forall y \in \Omega^Y, \forall t \in \Omega^T \quad (27)$$

$$\bar{h}^{po} \leq h_{d,y,t}^{po} \leq \underline{h}^{po}, h_{d,y,0}^{po} \leq h_{d,y,T}^{po}, \forall d \in \Omega^D, \forall y \in \Omega^Y, \forall t \in \Omega^T \quad (28)$$

$$E_{d,y,t} = E_{d,y,t-1} + \eta^{ch} P_{d,y,t}^{ch} - P_{d,y,t}^{dch} / \eta^{dch}, \forall d \in \Omega^D, \forall y \in \Omega^Y, \forall t \in \Omega^T \quad (29)$$

$$0 \leq P_{d,y,t}^{ch} \leq \bar{P}^{ch}, 0 \leq P_{d,y,t}^{dch} \leq \bar{P}^{dch}, \forall d \in \Omega^D, \forall y \in \Omega^Y, \forall t \in \Omega^T \quad (30)$$

$$SOC_d^{\min} E_d^{\text{rated}} \leq E_{d,y,t} \leq SOC_d^{\max} E_d^{\text{rated}}, E_{d,y,0} = E_{d,y,T}, \forall d \in \Omega^D, \forall y \in \Omega^Y, \forall t \in \Omega^T \quad (31)$$

Eqs. (27)–(28) represent the operational constraints of the IDC's tanks, specifically the hydraulic head balance constraint and the hydraulic head state constraint, respectively. Eqs. (29)–(31) define the operational constraints of the IDC's backup battery systems, including the energy balance constraint, the charging/discharging state constraint, and the capacity constraint.

We adopt a DDU set, as shown in Eqs. (32)–(35), to model the data load ($\bar{D}_{d,y,t}^0$) uncertainty.

$$\Xi(x_{d,y,t}^{ser}) = \left\{ \bar{D}_{d,y,t}^0 \geq 0 \mid \bar{D}_{d,y,t}^0 \leq \bar{\bar{D}}_{d,y,t}^0 + \bar{\alpha}_{d,y,t} x_{d,y,t}^{ser}, \forall d \in \Omega^D, \forall y \in \Omega^Y, \forall t \in \Omega^T \right\} \quad (32)$$

$$\bar{D}_{d,y,t}^0 \geq \bar{\bar{D}}_{d,y,t}^0 + \underline{\alpha}_{d,y,t} x_{d,y,t}^{ser}, \forall d \in \Omega^D, \forall y \in \Omega^Y, \forall t \in \Omega^T \quad (33)$$

$$\sum_{d \in \Omega^D} \bar{D}_{d,y,t}^0 \leq \bar{\bar{D}}_{y,t}^{\text{all}} + \bar{\beta}_{y,t} \sum_{d \in \Omega^D} x_{d,y,t}^{ser}, \forall y \in \Omega^Y, \forall t \in \Omega^T \quad (34)$$

$$\sum_{d \in \Omega^D} \bar{D}_{d,y,t}^0 \geq \bar{\bar{D}}_{y,t}^{\text{all}} + \underline{\beta}_{y,t} \sum_{d \in \Omega^D} x_{d,y,t}^{ser}, \forall y \in \Omega^Y, \forall t \in \Omega^T \quad (35)$$

Eqs. (32)–(33) respectively define the upper and lower bounds of the data load $\bar{D}_{d,y,t}^0$ within a region, where $\bar{\bar{D}}_{d,y,t}^0$ and $\bar{\bar{D}}_{d,y,t}^0$ represent the baseline boundaries. Induction coefficients $\bar{\alpha}_{d,y,t}$ and $\underline{\alpha}_{d,y,t}$ are introduced to capture the demand inducing effects within each region. $\bar{\beta}_{y,t}$ and $\underline{\beta}_{y,t}$ represent the demand coefficients of the total data load across the entire

region. The DDU set establishes a relationship between the data load and the number of active servers, such that the bounds of $\tilde{D}_{d,y,t}^0$ increase as $x_{d,y}^{\text{ser}}$ grows. Eqs. (34)–(35) provide the variation range of the total data load $\sum_{d \in \Omega^D} \tilde{D}_{d,y,t}^0$, which is also decision-dependent. Mathematically, these constraints help prevent over-conservatism in the DDU set formulation. The induction coefficients are estimated from historical IDC load data through statistical regression and sensitivity analysis, and their magnitudes are subsequently adjusted according to operational conditions.

3.2. Three-level model structure and objective function of IEWS

The first component of the IEWS objective function includes the reinforcement investment costs for distribution lines. The second component accounts for interaction costs related to electricity and water transactions with the IDC. The third part represents the operational cost under uncertainties in renewable energy sources (RES) output, power and water loads, and distribution network failures. The formulation of the first component of the objective function and its associated constraints are presented as follows:

$$F_{\text{IEWS}}^I = \sum_{y \in \Omega^Y} \frac{1}{(1 + \zeta)^{y-1}} \left(\sum_{(i,j) \in \Omega^L} c_{ij}^{\text{al}} A_{ij}^{\text{al}} + \sum_{g \in \Omega^G} c_g^{\text{ag}} A_{g,y}^{\text{ag}} \right) \quad (36)$$

$$A_{ij,y+1}^{\text{al}} = x_{ij,y+1}^{\text{al}} - x_{ij,y}^{\text{al}}, \forall (i,j) \in \Omega^L, \forall y \in \Omega^Y \quad (37)$$

$$A_{g,y+1}^{\text{ag}} = x_{g,y+1}^{\text{ag}} - x_{g,y}^{\text{ag}}, \forall g \in \Omega^G, \forall y \in \Omega^Y \quad (38)$$

$$x_{ij,y}^{\text{al}} \leq 1, x_{g,y}^{\text{ag}} \leq 1, \forall (i,j) \in \Omega^L, \forall g \in \Omega^G, \forall y \in \Omega^Y \quad (39)$$

Eq. (36) calculates the total investment cost for IEWS reinforcement, specifically summing the costs for line hardening and generator capacity upgrades. Eqs. (37)–(38) indicate whether a distribution line or generator unit has been reinforced. Eq. (39) imposes a constraint ensuring that each line and each piece of equipment can be reinforced only once.

The second component of the IEWS objective function, along with its associated constraints, is formulated as follows:

$$F_{\text{IEWS}}^{\text{II}} = \sum_{y \in \Omega^Y} \frac{N^y}{(1 + \zeta)^{y-1}} \sum_{t \in \Omega^T} \left(C_{y,t}^{\text{grid}} + C_{y,t}^{\text{hw}} - \sum_{d \in \Omega^D} (C_{d,y,t}^{\text{ele}} + C_{d,y,t}^{\text{wat}}) \right) \quad (40)$$

$$C_{y,t}^{\text{grid}} = c_{y,t}^{\text{grid}} p_{y,t}^{\text{grid}} \Delta t, \forall y \in \Omega^Y, \forall t \in \Omega^T \quad (41)$$

$$C_{y,t}^{\text{hw}} = c_{y,t}^{\text{hw}} f_{y,t}^{\text{hw}} \Delta t, \forall y \in \Omega^Y, \forall t \in \Omega^T \quad (42)$$

$$C_{d,y,t}^{\text{ele}} = c_{d,y,t}^{\text{ele}} p_{d,y,t}^{\text{DC}} \Delta t, \forall d \in \Omega^D, \forall y \in \Omega^Y, \forall t \in \Omega^T \quad (43)$$

$$C_{d,y,t}^{\text{wat}} = c_{d,y,t}^{\text{wat}} f_{d,y,t}^{\text{DC}} \Delta t, \forall d \in \Omega^D, \forall y \in \Omega^Y, \forall t \in \Omega^T \quad (44)$$

Eq. (40) represents the energy trading cost, reflecting the electricity and water exchange expenses with the upstream grid/source and the IDC. Eqs. (41)–(42) represent the costs of purchasing electricity from the main power grid and water from water sources, respectively. Eqs. (43)–(44) capture the basic interaction costs with the IDC.

The third component of the objective function is formulated as follows:

$$F_{\text{IEWS}}^{\text{III}} = \sum_{y \in \Omega^Y} \frac{N^y}{(1 + \zeta)^{y-1}} \sum_{i \in \Omega^I} \sum_{t \in \Omega^T} (c^{e,\text{curt}} p_{i,y,t}^{\text{curt}} + c^{w,\text{curt}} f_{w,y,t}^{\text{curt}}) \Delta t \quad (45)$$

Eq. (45) defines the operational penalty cost, quantifying the economic loss associated with shedding electrical and water loads under worst-case scenarios. The optimization is subject to the following technical constraints:

(1) Power network constraints. The constraints ensuring nodal power balance and transmission safety limits are defined as follows:

$$\sum_{j \in \Gamma(i)} p_{ji,y,t} + p_{i,y,t}^{\text{res}} + p_{i,y,t}^{\text{gen}} + p_{i,y,t}^{\text{grid}} = \sum_{k \in \Lambda(i)} p_{ik,t,s} + p_{i,y,t}^{\text{DC}} + p_{i,y,t}^{\text{lwp}} + \tilde{p}_{i,y,t}^{\text{L}}, \forall i \in \Omega^I, \forall y \in \Omega^Y, \forall t \in \Omega^T \quad (46)$$

$$\sum_{j \in \Gamma(i)} Q_{ji,y,t} + Q_{i,y,t}^{\text{RES}} + Q_{i,y,t}^{\text{gen}} = \sum_{k \in \Lambda(i)} Q_{ik,t,s} + Q_{i,y,t}^{\text{DC}} + Q_{i,y,t}^{\text{lwp}} + Q_{i,y,t}^{\text{L}}, \forall i \in \Omega^I, \forall y \in \Omega^Y, \forall t \in \Omega^T \quad (47)$$

$$V_{i,y,t} - V_{j,y,t} = 2p_{ij,y,t} R_{ij}^L + 2Q_{ij,t,s} X_{ij}^L, \forall i \in \Omega^I, \forall y \in \Omega^Y, \forall t \in \Omega^T \quad (48)$$

$$\tilde{u}_{ij,y,t} \left(p_{ij}^{\text{max}} + x_{g,y}^{\text{al}} p_{ij}^{\text{al}} \right) \leq p_{ij,y,t} \leq \tilde{u}_{ij,y,t} \left(p_{ij}^{\text{max}} + x_{g,y}^{\text{al}} p_{ij}^{\text{al}} \right), \forall (i,j) \in \Omega^L, \forall y \in \Omega^Y, \forall t \in \Omega^T \quad (49)$$

$$\tilde{u}_{ij,y,t} \left(Q_{ij}^{\text{max}} + x_{g,y}^{\text{al}} Q_{ij}^{\text{al}} \right) \leq Q_{ij,y,t} \leq \tilde{u}_{ij,y,t} \left(Q_{ij}^{\text{max}} + x_{g,y}^{\text{al}} Q_{ij}^{\text{al}} \right), \forall (i,j) \in \Omega^L, \forall y \in \Omega^Y, \forall t \in \Omega^T \quad (50)$$

$$\underline{V} \leq V_{i,y,t} \leq \bar{V}, \forall i \in \Omega^I, \forall y \in \Omega^Y, \forall t \in \Omega^T \quad (51)$$

$$\tilde{v}_{g,y,t} \left(p_g^{\text{gen}} + x_{g,y}^{\text{ag}} p_g^{\text{ag}} \right) \leq p_{g,y,t}^{\text{gen}} \leq \tilde{v}_{g,y,t} \left(p_g^{\text{gen}} + x_{g,y}^{\text{ag}} p_g^{\text{ag}} \right), \forall g \in \Omega^G, \forall y \in \Omega^Y, \forall t \in \Omega^T \quad (52)$$

$$0 \leq p_{i,y,t}^{\text{res}} \leq \tilde{p}_{i,y,t}^{\text{res}}, \forall i \in \Omega^I, \forall y \in \Omega^Y, \forall t \in \Omega^T \quad (53)$$

$$\tilde{v}_{g,y,t} \left(p_g^{\text{gen}} + x_{g,y}^{\text{ag}} p_g^{\text{ag}} \right) \tan \underline{\theta} \leq Q_{g,y,t}^{\text{gen}} \leq \tilde{v}_{g,y,t} \left(p_g^{\text{gen}} + x_{g,y}^{\text{ag}} p_g^{\text{ag}} \right) \tan \bar{\theta}, \forall g \in \Omega^G, \forall y \in \Omega^Y, \forall t \in \Omega^T \quad (54)$$

Eq. (46) represents the active power balance at each node, while Eq. (47) captures the reactive power balance. Eq. (48) formulates the power flow relations for distribution lines. Eqs. (49)–(50) impose constraints on the active and reactive power flows of each branch, ensuring they do not exceed maximum transmission capacities, with fault conditions duly considered. Eq. (51) constrains the squared voltage magnitude at each node within permissible limits. Eq. (52) ensures that generator outputs remain within predefined bounds and are governed by their respective fault states. Eq. (53) constrains the output of renewable energy sources (RES), ensuring that it does not exceed their maximum generation capacity. Eq. (54) constrains generator reactive power output based on power factor angle limits.

(2) Water network constraints. The hydraulic constraints describing water flow dynamics and pressure requirements are formulated as follows:

$$\underline{h}_{p,y,t}^{\Delta,\text{ini}} \leq h_{p,y,t}^{\Delta,\text{ini}} \leq \bar{h}_p^{\Delta,\text{ini}}, \underline{h}_{p,y,t}^{\Delta,\text{ter}} \leq h_{p,y,t}^{\Delta,\text{ter}} \leq \bar{h}_p^{\Delta,\text{ter}}, \Delta \in \{\text{lw}, \text{lwp}\}, \forall p \in \Omega^P, \forall y \in \Omega^Y, \forall t \in \Omega^T \quad (55)$$

$$h_{p,y,t}^{\Delta,\text{hl}} = \left(h_{p,y,t}^{\Delta,\text{ini}} + \bar{h}_{p,y,t}^{\Delta,\text{ini}} \right) - \left(h_{p,y,t}^{\Delta,\text{ter}} + \bar{h}_{p,y,t}^{\Delta,\text{ter}} \right), \Delta \in \{\text{lw}, \text{lwp}\}, \forall p \in \Omega^P, \forall y \in \Omega^Y, \forall t \in \Omega^T \quad (56)$$

$$h_{p,y,t}^{\Delta,\text{hl}} \geq 0, \Delta \in \{\text{lw}, \text{lwp}\}, \forall p \in \Omega^P, \forall y \in \Omega^Y, \forall t \in \Omega^T \quad (57)$$

$$h_{p,y,t}^{\text{lwp,hl}} + a_p^{\text{lwp}} f_{p,y,t}^{\text{lwp}} + b_p^{\text{lwp}} = R_p^{\text{lwp}} \left(f_{p,y,t}^{\text{lwp}} \right)^2, \forall p \in \Omega^P, \forall y \in \Omega^Y, \forall t \in \Omega^T \quad (58)$$

$$h_{p,y,t}^{\text{lw,hl}} = R_p^{\text{lw}} \left(f_{p,y,t}^{\text{lw}} \right)^2, \forall p \in \Omega^P, \forall y \in \Omega^Y, \forall t \in \Omega^T \quad (59)$$

$$0 \leq f_{p,y,t}^{\Delta} \leq f_p^{\Delta, \max}, \Delta \in \{lw, lwp\}, \forall p \in \Omega^p, \forall y \in \Omega^y, \forall t \in \Omega^t \quad (60)$$

$$p_{p,y,t}^{lwp} = \left(a_p^{lwp} (f_{p,y,t}^{lwp})^2 + b_p^{lwp} f_{p,y,t}^{lwp} \right) / \pi_p^{lwp}, \forall p \in \Omega^p, \forall y \in \Omega^y, \forall t \in \Omega^t \quad (61)$$

$$f_{w,y,t}^{hw} + \sum_{w \in \{ter\}} f_{p,y,t}^{\Delta} - \sum_{w \in \{ini\}} f_{p,y,t}^{\Delta} + f_{w,y,t}^{curt} = f_{w,y,t}^{DC} + \tilde{f}_{w,y,t}^L, \Delta \in \{lw, lwp\}, \forall w \in \Omega^w, \forall p \in \Omega^p, \forall y \in \Omega^y, \forall t \in \Omega^t \quad (62)$$

Eqs. (55)–(57) model the hydraulic characteristics of pipelines based on the Darcy-Weisbach equation. Eq. (55) imposes head constraints for pipelines with and without pumps. Eq. (56) calculates head loss for both types of pipelines. Eq. (57) ensures that head loss remains non-negative. Eq. (58) defines the coupling between head loss and flow for pipelines with pumps, whereas Eq. (59) does the same for pipelines without pumps. Eq. (60) constrains the flow within each pipeline to operate within allowable limits. Eq. (61) establishes the relationship between pump power consumption and water flow. Finally, Eq. (62) enforces mass balance constraints on water quantity at each node.

Remark 2. To facilitate tractable solution of the model, the bilinear structures appearing in Eqs. (49), (50), (52) and (54) can be linearized using the Big-M method. For example, a bilinear term of the form $y = ux$, $u \in \{0, 1\}$ can be equivalently transformed into the following set of linear constraints:

$$\begin{cases} y \leq Mu \\ y \leq x \\ y \geq x - M(1 - u) \\ y \geq 0 \end{cases} \quad (63)$$

Eqs. (58) and (61) adopt the two-step solution method proposed in [36], which approximates the nonlinear and non-convex model using Mixed-Integer Second-Order Cone Programming, followed by feasibility recovery via the Penalty Convex-Concave Procedure.

We adopt a DIU set to model the uncertain outputs of RES ($\tilde{P}_{i,y,t}^{res}$), electricity load ($\tilde{P}_{i,y,t}^L$), and water load ($\tilde{f}_{w,y,t}^L$).

$$\mathbb{L}^{res} = \left\{ \begin{array}{l} \tilde{P}_{i,y,t}^{res} = \bar{P}_{i,y,t}^{res} + \Delta \tilde{P}_{i,y,t}^{res} u_{i,y,t}^{res,+} - \Delta \tilde{P}_{i,y,t}^{res} u_{i,y,t}^{res,-} \\ \sum_{t \in \Omega^t} (u_{i,y,t}^{res,+} + u_{i,y,t}^{res,-}) \leq \Gamma_{i,y}^{res} \\ u_{i,y,t}^{res,+} + u_{i,y,t}^{res,-} \leq 1, u_{i,y,t}^{res,+}, u_{i,y,t}^{res,-} \in \{0, 1\} \end{array} \right\} \quad (64)$$

$$\mathbb{L}^L = \left\{ \begin{array}{l} \tilde{P}_{i,y,t}^L = \bar{P}_{i,y,t}^L + \Delta \tilde{P}_{i,y,t}^L u_{i,y,t}^{el,+} - \Delta \tilde{P}_{i,y,t}^L u_{i,y,t}^{el,-} \\ \sum_{t \in \Omega^t} (u_{i,y,t}^{el,+} + u_{i,y,t}^{el,-}) \leq \Gamma_{i,y}^{el} \\ u_{i,y,t}^{el,+} + u_{i,y,t}^{el,-} \leq 1, u_{i,y,t}^{el,+}, u_{i,y,t}^{el,-} \in \{0, 1\} \end{array} \right\} \quad (65)$$

$$\mathbb{L}^{wl} = \left\{ \begin{array}{l} \tilde{f}_{w,y,t}^L = \bar{f}_{w,y,t}^L + \Delta \tilde{f}_{w,y,t}^L u_{w,y,t}^{wl,+} - \Delta \tilde{f}_{w,y,t}^L u_{w,y,t}^{wl,-} \\ \sum_{t \in \Omega^t} (u_{w,y,t}^{wl,+} + u_{w,y,t}^{wl,-}) \leq \Gamma_{w,y}^{wl} \\ u_{w,y,t}^{wl,+} + u_{w,y,t}^{wl,-} \leq 1, u_{w,y,t}^{wl,+}, u_{w,y,t}^{wl,-} \in \{0, 1\} \end{array} \right\} \quad (66)$$

We adopt a DDU set to model the uncertainties associated with distribution line faults ($\tilde{u}_{ij,y,t}$) and generator outages ($\tilde{v}_{g,y,t}$).

$$\theta(x_{ij,y}^{al}) = \left\{ \tilde{u}_{ij,y,t} \in \{0, 1\} \mid \sum_{t \in \Omega^t} \sum_{(i,j) \in \Omega^L} (1 - \tilde{u}_{ij,y,t}) \leq \Gamma_y^{al}, \forall y \in \Omega^y \right\} \quad (67)$$

$$\tilde{u}_{ij,y,t} \geq x_{ij,y}^{al}, \forall (i,j) \in \Omega^L, \forall y \in \Omega^y, \forall t \in \Omega^t \quad (68)$$

$$\theta(x_{g,y}^{ag}) = \left\{ \tilde{v}_{g,y,t} \in \{0, 1\} \mid \sum_{t \in \Omega^t} \sum_{g \in \Omega^G} (1 - \tilde{v}_{g,y,t}) \leq \Gamma_y^{ag}, \forall y \in \Omega^y \right\} \quad (69)$$

$$\tilde{v}_{g,y,t} \geq x_{g,y}^{ag}, \forall g \in \Omega^G, \forall y \in \Omega^y, \forall t \in \Omega^t \quad (70)$$

According to the N-k criterion, the total number of failed distribution lines and distributed generators (DGs) is constrained by Eqs. (67) and (69), respectively. In addition, proactive reinforcement decisions impose limits on $\tilde{u}_{ij,y,t}$ and $\tilde{v}_{g,y,t}$, reflecting decision-dependent relationships as formulated in Eqs. (68) and (70). By imposing the availability lower bounds $\tilde{u}_{ij,y,t} \geq x_{ij,y}^{al}$ and $\tilde{v}_{g,y,t} \geq x_{g,y}^{ag}$, any proactive reinforcement decision strictly forces the corresponding component to remain in the operational state (i.e., availability is locked at 1), thereby eliminating the possibility of failure for that specific asset. This mechanism endogenously shrinks the feasible region of the worst-case fault set as investment increases, preventing the adversary from attacking reinforced infrastructure. Γ_y^{al} and Γ_y^{ag} represent the maximum numbers of faults of transmission lines and generators during the operation period.

4. Design of Distributed Solution Method

The proposed solution methodology integrates three distinct algorithms including PC&CG, ATC, and AOP into a cohesive framework. It is crucial to clarify that this hybrid design is driven by the necessity to address three decoupled challenges that cannot be solved by a single existing method. First, the inclusion of DDU renders the uncertainty set dynamic and the robust subproblem bilinear. Standard C&CG fails to handle such structures efficiently; therefore, PC&CG is employed specifically to decompose the complex DDU space into tractable subproblems, ensuring iterative convergence to the global optimum of the DDU-ARP subproblem. Second, due to the strict data privacy barriers between IDC and IEWS stakeholders, centralized optimization is infeasible. ATC is selected as the decoupling mechanism to enable distributed computing via limited boundary information exchange. Third, since the planning model involves binary investment variables, the resulting problem is non-convex for which standard ATC lacks convergence guarantees. The AOP is thus integrated as an outer-loop strategy to iteratively handle integer variables, ensuring the stability and convergence of the overall distributed framework.

4.1. PC&CG algorithm for three-level robust optimization with DDU

For the sake of brevity and without loss of generality, the planning problems of the IDC and IEWS are mathematically reformulated in matrix form as the following DDU-ARP model:

$$F_k = \min_{\substack{x_k \in X_k \\ q_k \in Q_k(x_k)}} c_k^T x_k + e_k^T q_k + \max_{\substack{\zeta_k \in \Pi(x_k) \\ \varpi_k \in \Gamma_k}} \min_{y_k \in Y_k(x_k, q_k, \zeta_k)} d_k^T y_k, k \in \{DC, IEWS\} \quad (71)$$

$$X_k = \{x_k \in \{0, 1\}^{n_{x,k}^1} \times \mathbb{N}^{n_{x,k}^2} \mid A_k x_k \leq b_k\}, k \in \{DC, IEWS\} \quad (72)$$

$$Q_k(x_k) = \{q_k \in \mathbb{R}_+^{n_{q,k}} \mid G_k q_k \leq g_k - H_k x_k\}, k \in \{DC, IEWS\} \quad (73)$$

$$\Pi_k(x_k) = \{\zeta_k \in \{0, 1\}^{n_{\zeta,k}^1} \times \mathbb{R}_+^{n_{\zeta,k}^2} \mid I_k \zeta_k \leq i_k - J_k x_k\}, k \in \{DC, IEWS\} \quad (74)$$

$$\begin{aligned} \|M_{ck}y_k + m_{ck1}\|_2 &\leq m_{ck2} + n_{ck}^T y_k - o_{ck}^T \hat{x}_k - p_{ck}^T \hat{q}_k - q_{ck}^T \hat{\zeta}_k - r_{ck}^T \hat{\varpi}_k, \hat{u}_{ck} \\ &\in \hat{\mathcal{P}}_k(u_{ck}), k \in \{\text{DC, IEWS}\} \end{aligned} \quad (88)$$

$$(\zeta_k^{\hat{u}_{ck}}, \lambda_k^{\hat{u}_{ck}}) \in \mathcal{U}_k(\mathbf{x}_k, u_{ck}), k \in \{\text{DC, IEWS}\} \quad (89)$$

$$\mathbf{x}_k \in \{0, 1\}^{n_{xk1}} \times \mathbb{N}^{n_{xk2}}, k \in \{\text{DC, IEWS}\} \quad (90)$$

Where, $\hat{\mathcal{P}}(u_{ck})$ denotes the set of identified extreme points u_{ck} accumulated up to the current iteration.

Solving **MP** to obtain the optimal solution $(\hat{\mathbf{x}}_k, \hat{\mathbf{q}}_k, \hat{\eta}_k)$, this provides a lower bound for the DDU-ARP:

$$LB_k = \mathbf{c}_k^T \hat{\mathbf{x}}_k + \mathbf{e}_k^T \hat{\mathbf{q}}_k + \hat{\eta}_k, k \in \{\text{DC, IEWS}\} \quad (91)$$

Using the upper-level solution derived **MP**, the subproblem (**SP**) is defined as follows:

$$SP_k : \max_{\substack{\zeta_k \in \Pi(\hat{\mathbf{x}}_k) \\ \varpi_k \in \hat{\mathcal{V}}_k}} \min_{y_k \in Y_k(\hat{\mathbf{x}}_k, \hat{\mathbf{q}}_k, \hat{\zeta}_k)} d_k^T y_k, k \in \{\text{DC, IEWS}\} \quad (92)$$

Since full recourse is assumed, **SP** is always feasible. Furthermore, **SP** originally in a max-min form, can be transformed into a single-level problem by applying the KKT conditions of its inner minimization problem, as follows:

$$SP_k : \max d_k^T y_k, k \in \{\text{DC, IEWS}\} \quad (93)$$

$$I_k \zeta_k \leq \mathbf{i}_k - J_k \hat{\mathbf{x}}_k, k \in \{\text{DC, IEWS}\} \quad (94)$$

$$K_k \varpi_k \leq \mathbf{l}_k, k \in \{\text{DC, IEWS}\} \quad (95)$$

$$\begin{aligned} u_{c2k}^T (M_{ck} y_k + m_{ck1}) + u_{c1k} (m_{ck2} - o_{ck}^T \hat{\mathbf{x}}_k - p_{ck}^T \hat{\mathbf{q}}_k - q_{ck}^T \hat{\zeta}_k - r_{ck}^T \hat{\varpi}_k) &= 0, \forall c \\ &\in \Omega^C, k \in \{\text{DC, IEWS}\} \end{aligned} \quad (99)$$

Here, Eqs. (96)–(97) represent the primal and dual constraints of the lower-level problem in the SP, respectively; Eq. (98) corresponds to the stationarity condition; and Eq. (99) denotes the complementarity constraints. The bilinear terms in Eq. (99) can be linearized using the Big-M method, as described in Eq. (63).

By solving **SP**, the optimal lower-level variables and their corresponding extreme point, i.e., $(y_k, \hat{\mathbf{u}}_{1k}, \hat{\mathbf{u}}_{2k}, \dots, \hat{\mathbf{u}}_{ck})$, are obtained. Subsequently, we update $\hat{\mathcal{P}}_k \leftarrow \hat{\mathcal{P}}_k \cup \{\hat{\mathbf{u}}_{ck}\}$, $\forall c \in \Omega^C$, introduce new variables $y_k^{\hat{u}_{ck}}, \zeta_k^{\hat{u}_{ck}}, \varpi_k^{\hat{u}_{ck}}$ and $\lambda_k^{\hat{u}_{ck}}$, and add the new constraints (100)–(102) as optimality cuts to **MP** in order to strengthen it.

$$\eta_k \geq d_k^T y_k^{\hat{u}_{ck}}, y_k^{\hat{u}_{ck}} \in \mathbb{R}_+^{n_{y_k}} \quad (100)$$

$$\|M_{ck} y_k^{\hat{u}_{ck}} + m_{ck1}\|_2 \leq m_{ck2} + n_{ck}^T y_k^{\hat{u}_{ck}} - o_{ck}^T \hat{\mathbf{x}}_k - p_{ck}^T \hat{\mathbf{q}}_k - q_{ck}^T \hat{\zeta}_k - r_{ck}^T \hat{\varpi}_k \quad (101)$$

$$(\zeta_k^{\hat{u}_{ck}}, \lambda_k^{\hat{u}_{ck}}) \in \mathcal{U}_k(\mathbf{x}_k, \hat{u}_{ck}) \quad (102)$$

Meanwhile, since the solution obtained from solving **MP** and **SP** constitutes a feasible solution to the DDU-ARP, the upper bound (UB) of the DDU-ARP can be updated accordingly:

$$UB_k = \min\{UB_k, \mathbf{c}_k^T \hat{\mathbf{x}}_k + \mathbf{e}_k^T \hat{\mathbf{q}}_k + d_k^T \hat{\mathbf{y}}_k\}, k \in \{\text{DC, IEWS}\} \quad (103)$$

Algorithm 1 outlines the overall procedure of the PC&CG method, and Ref. [40] proves the finite convergence of the algorithm.

Algorithm 1. PC&CG for IDC/IEWS.

Algorithm 1: PC&CG for IDC/IEWS

Step 1: Initialize

Set $LB_k = -\infty$, $UB_k = +\infty$, $\varepsilon_1 = 0$;

Initialize solution set $\hat{\mathcal{P}}_k$.

Step 2: Repeat until convergence:

While $(UB_k - LB_k) / |LB_k| > \varepsilon_1$:

2.1 Solve **MP**:

If **MP** is infeasible:

Terminate and report infeasibility.

Else:

Let $(\hat{\mathbf{x}}_k, \hat{\mathbf{q}}_k, \hat{\eta}_k)$ be optimal solution.

Update $LB_k = \mathbf{c}_k^T \hat{\mathbf{x}}_k + \mathbf{e}_k^T \hat{\mathbf{q}}_k + \hat{\eta}_k$

2.2 Solve **SP**:

Get optimal solution $(y_k, \hat{\mathbf{u}}_{1k}, \hat{\mathbf{u}}_{2k}, \dots, \hat{\mathbf{u}}_{ck})$

Update $\hat{\mathcal{P}}_k \leftarrow \hat{\mathcal{P}}_k \cup \{\hat{\mathbf{u}}_{ck}\}, \forall c \in \Omega^C$

Add $y_k^{\hat{u}_{ck}}, \zeta_k^{\hat{u}_{ck}}, \varpi_k^{\hat{u}_{ck}}, \lambda_k^{\hat{u}_{ck}}$ and constraints (100)–(102) to **MP**.

2.3 Update UB:

$$UB_k = \min\{UB_k, \mathbf{c}_k^T \hat{\mathbf{x}}_k + \mathbf{e}_k^T \hat{\mathbf{q}}_k + d_k^T \hat{\mathbf{y}}_k\}$$

Step 3: Return $(\hat{\mathbf{x}}_k, \hat{\mathbf{q}}_k)$ as the solution to the original problem.

$$\begin{aligned} \|M_{ck}y_k + m_{ck1}\|_2 &\leq m_{ck2} + n_{ck}^T y_k - o_{ck}^T \hat{\mathbf{x}}_k - p_{ck}^T \hat{\mathbf{q}}_k - q_{ck}^T \hat{\zeta}_k - r_{ck}^T \hat{\varpi}_k, \forall c \in \Omega^C, k \\ &\in \{\text{DC, IEWS}\} \end{aligned} \quad (96)$$

$$\|u_{c2k}\| \leq u_{c1k}, \forall c \in \Omega^C, k \in \{\text{DC, IEWS}\} \quad (97)$$

$$d_k + \sum_{c \in \Omega^C} M_{ck}^T u_{c2k} + u_{c1k} n_{ck} = 0, \forall c \in \Omega^C, k \in \{\text{DC, IEWS}\} \quad (98)$$

4.2. ATC-based multi-agent coordination method

As a decentralized optimization algorithm, the ATC method is based on the principle of decomposing the entire system into multiple subsystems and incorporating penalty terms into each subsystem's objective function. To ensure the consistency of the coupling variables (electricity and water exchange quantities) between the IDC and IEWS, the Augmented Lagrangian penalty terms are explicitly added to the interaction cost objectives defined in Eq. (5) and Eq. (40) during the iterative

solution process. The specific formulations of these penalty functions are as follows:

$$C_{d,y,t}^{\text{dcpe}} = \left(\gamma_{d,y,t}^{\text{DC-ele}} (P_{d,y,t}^{\text{DC}} - \hat{P}_{d,y,t}^{\text{DC}}) + \rho^{\text{ele}} \|P_{d,y,t}^{\text{DC}} - \hat{P}_{d,y,t}^{\text{DC}}\|_2^2 \right) \Delta t + \left(\gamma_{d,y,t}^{\text{DC-wat}} (f_{d,y,t}^{\text{DC}} - \hat{f}_{d,y,t}^{\text{DC}}) + \rho^{\text{wat}} \|f_{d,y,t}^{\text{DC}} - \hat{f}_{d,y,t}^{\text{DC}}\|_2^2 \right) \Delta t \quad (104)$$

$$\forall d \in \Omega^D, \forall y \in \Omega^Y, \forall t \in \Omega^T$$

$$C_{d,y,t}^{\text{ewpe}} = \left(\gamma_{d,y,t}^{\text{IEWS-ele}} (P_{d,y,t}^{\text{DC}} - \hat{P}_{d,y,t}^{\text{DC}}) + \rho^{\text{ele}} \|P_{d,y,t}^{\text{DC}} - \hat{P}_{d,y,t}^{\text{DC}}\|_2^2 \right) \Delta t + \left(\gamma_{d,y,t}^{\text{IEWS-wat}} (f_{d,y,t}^{\text{DC}} - \hat{f}_{d,y,t}^{\text{DC}}) + \rho^{\text{wat}} \|f_{d,y,t}^{\text{DC}} - \hat{f}_{d,y,t}^{\text{DC}}\|_2^2 \right) \Delta t \quad (105)$$

$$\forall d \in \Omega^D, \forall y \in \Omega^Y, \forall t \in \Omega^T$$

First, the system is partitioned into the IDC and IEWS based on the interests of different stakeholders. Then, considering the interdependent exchange of power and water supply, the IDC and IEWS optimize their respective objectives to maximize individual benefits while integrating penalty functions that guide them toward a joint agreement. In the v th iteration, the interactive power/water supply optimized by the IDC and IEWS are $(P_{d,y,t}^{\text{DC}(v)}, P_{d,y,t}^{\text{DC}(v)})$ and $(f_{d,y,t}^{\text{DC}(v)}, f_{d,y,t}^{\text{DC}(v)})$, respectively. These values are then updated and fed back to the IDC and IEWS. The new interactive power/water supply is given as follows:

$$\hat{P}_{d,y,t}^{\text{DC}(v+1)} = (P_{d,y,t}^{\text{DC}(v)} + P_{d,y,t}^{\text{DC}(v)}) / 2 \quad (106)$$

$$\hat{f}_{d,y,t}^{\text{DC}(v+1)} = (f_{d,y,t}^{\text{DC}(v)} + f_{d,y,t}^{\text{DC}(v)}) / 2 \quad (107)$$

$(\gamma_{d,y,t}^{\text{DC-ele}}, \gamma_{d,y,t}^{\text{DC-wat}}, \gamma_{d,y,t}^{\text{IEWS-ele}}, \gamma_{d,y,t}^{\text{IEWS-wat}})$ and $(\rho^{\text{ele}}, \rho^{\text{wat}})$ represent the linear and quadratic multipliers of the Lagrangian penalty function, respectively. These multipliers are used to relax the system consistency con-

straints and ensure the feasibility of the subproblems. The initial values of the linear multipliers γ can be set close to zero, while the quadratic

multipliers ρ are typically initialized within the range of 0 to 2. To ensure convergence, the multipliers are updated after each iteration as follows [41]:

$$\begin{cases} \gamma_{d,y,t}^{\text{DC-ele}(v+1)} = \gamma_{d,y,t}^{\text{DC-ele}(v)} + 2(\rho^{\text{DC-ele}(v)})^2 (P_{d,y,t}^{\text{DC}(v)} - \hat{P}_{d,y,t}^{\text{DC}(v)}) \\ \rho^{\text{DC-ele}(v+1)} = \beta \rho^{\text{DC-ele}(v)} \end{cases} \quad (108)$$

$$\begin{cases} \gamma_{d,y,t}^{\text{IEWS-ele}(v+1)} = \gamma_{d,y,t}^{\text{IEWS-ele}(v)} + 2(\rho^{\text{IEWS-ele}(v)})^2 (P_{d,y,t}^{\text{DC}(v)} - \hat{P}_{d,y,t}^{\text{DC}(v)}) \\ \rho^{\text{IEWS-ele}(v+1)} = \beta \rho^{\text{IEWS-ele}(v)} \end{cases} \quad (109)$$

$$\begin{cases} \gamma_{d,y,t}^{\text{DC-wat}(v+1)} = \gamma_{d,y,t}^{\text{DC-wat}(v)} + 2(\rho^{\text{DC-wat}(v)})^2 (f_{d,y,t}^{\text{DC}(v)} - \hat{f}_{d,y,t}^{\text{DC}(v)}) \\ \rho^{\text{DC-wat}(v+1)} = \beta \rho^{\text{DC-wat}(v)} \end{cases} \quad (110)$$

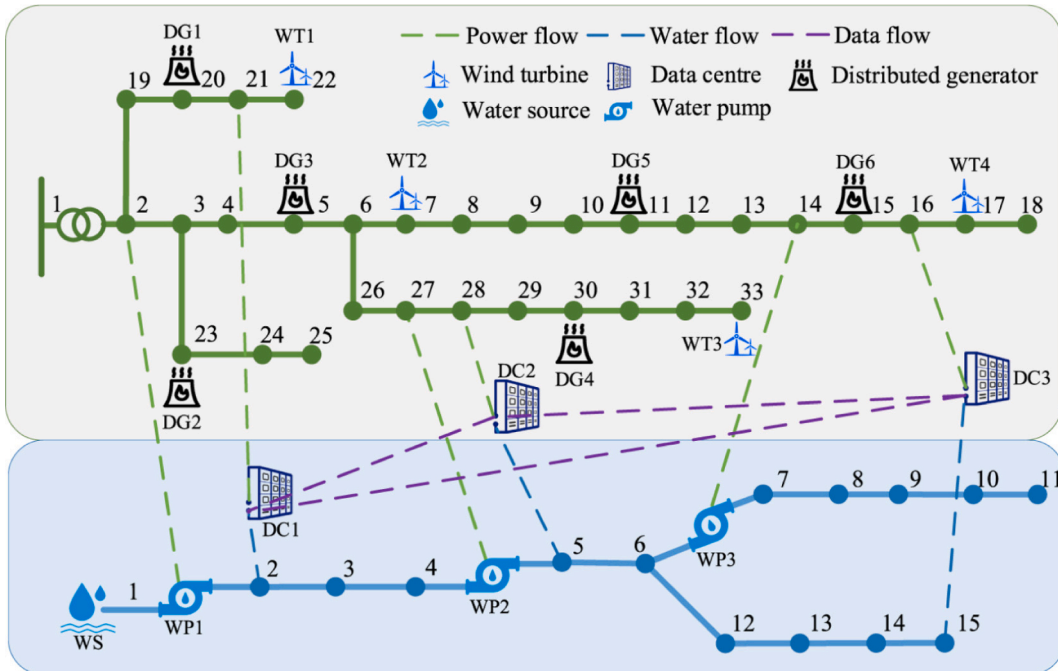


Fig. 3. Tested system.

$$\begin{cases} \gamma_{d,y,t}^{\text{IEWS-wat}(v+1)} = \gamma_{d,y,t}^{\text{IEWS-wat}(v)} + 2(\rho^{\text{IEWS-wat}(v)})^2 (f_{d,y,t}^{\text{DC}(v)} - \hat{f}_{d,y,t}^{\text{DC}(v)}) \\ \rho^{\text{IEWS-wat}(v+1)} = \beta \rho^{\text{IEWS-wat}(v)} \end{cases} \quad (111)$$

Using the updated interactive power/water supply values and Lagrangian multipliers, the IDC and IEWS are re-optimized. The convergence criteria are defined as follows:

$$\max \left\{ \left| \frac{P_{d,y,t}^{\text{DC}} - \hat{P}_{d,y,t}^{\text{DC}}}{\hat{P}_{d,y,t}^{\text{DC}}} \right|, \left| \frac{P_{d,y,t}^{\text{DC}} - \hat{P}_{d,y,t}^{\text{DC}}}{\hat{P}_{d,y,t}^{\text{DC}}} \right|, \left| \frac{f_{d,y,t}^{\text{DC}} - \hat{f}_{d,y,t}^{\text{DC}}}{\hat{f}_{d,y,t}^{\text{DC}}} \right|, \left| \frac{f_{d,y,t}^{\text{DC}} - \hat{f}_{d,y,t}^{\text{DC}}}{\hat{f}_{d,y,t}^{\text{DC}}} \right| \right\} \leq \varepsilon_2 \quad (112)$$

The convergence of the ATC-based coordination has been theoretically proved in [37].

Algorithm 2 presents the overall procedure of the ATC method.

Algorithm 2. ATC for IDC and IEWS.

Algorithm 2: ATC for IDC and IEWS

Initialization:

Given integer variables $\hat{\mathbf{x}}_k$ for IDC and IEWS;

Set initial multipliers $\gamma^{(0)}$;

Set tolerance $\varepsilon_2 = 10^{-3}$ and iteration index $v = 0$.

Repeat:

Step 1:

Update interactive power/water supply values using Eq. (106)-(107).

Step 2:

For IDC and IEWS:

Solve for optimal power ($P_{d,y,t}^{\text{DC}(v)}$, $P_{d,y,t}^{\text{DC}(v)}$) and ($f_{d,y,t}^{\text{DC}(v)}$, $f_{d,y,t}^{\text{DC}(v)}$) with ($\gamma_{d,y,t}^{\text{DC-elic}}$, $\gamma_{d,y,t}^{\text{DC-wat}}$, $\gamma_{d,y,t}^{\text{IEWS-elic}}$, $\gamma_{d,y,t}^{\text{IEWS-wat}}$) and (ρ^{elic} , ρ^{wat}) using Algorithm 1.

Step 3:

For IDC and IEWS:

Check convergence (Eq. (112));

If converged: output result and stop;

Else: go to Step 4.

Step 4:

Update multipliers using Eq. (108);

Set $v = v + 1$; return to Step 1.

Until convergence.

Remark 4. With the integer variables $\hat{\mathbf{x}}_k$ fixed, the planning problems for both the IDC and IEWS at this stage are converted into conventional DIU-type three-level robust optimization problems, which can likewise be solved using Algorithm 1.

4.3. AOP method for solving non-convex problems

Due to the presence of integer variables in the SDRP model, the problem is non-convex, and the standard ATC algorithm cannot guarantee convergence. To address this, the AOP method is employed to decompose the original problem into a q -fixed problem (with fixed interactive power/water supply) and an $\mathbf{x}$ -fixed problem (with fixed investment variables), which are solved separately. First, based on the

Table 3
Parameters of IDC.

Parameter	Value	Parameter	Value
T^{Delay}	0.5 s	η^{lo}	2.2568 kJ/m ³
$\chi^{\text{DL}}/\chi^{\text{RL}}$	0.2/0.8	μ_1/μ_2	0.2857/0.15
$p^{\text{ser-p}}/p^{\text{ser-i}}$	600/300 W	$T^{\text{in}}/\bar{T}^{\text{in}}$	20/25 °C
$\eta^{\text{ch}}/\eta^{\text{dch}}$	0.95	$c^{\text{d,curt}}$	3.3 million/TB
ϕ	0.05	c^{DR}	0.05 million/TBs
η^{ser}	0.85	ψ	500 request/s
$SOC^{\text{min}}/SOC^{\text{max}}$	0.3/0.9	R_d	3.5 °C/kW

initial settings of the linear and quadratic Lagrangian multipliers, the relaxed problems for IDC and IEWS with penalty functions are solved under the ATC decentralized framework to obtain the initial interactive power/water supply values. Second, using these initial values, the q -fixed problem is solved using the Algorithm 1 to optimize the integer investment variables $\mathbf{x}$ for equipment and line configurations within IDC

and IEWS. Third, with the optimized integer variables $\mathbf{x}$, the $\mathbf{x}$ -fixed problem is solved using Algorithm 2 to update the interactive power/water supply. In this iterative process, the integer variables $\mathbf{x}$ are taken as the final solution, and convergence is considered achieved when their values remain unchanged from the previous iteration. The detailed procedure and convergence proof can be found in [42]. Algorithm 3 summarizes the complete AOP-based alternating optimization framework, integrating Algorithm 1 and Algorithm 2 procedures with unified multiplier updates and convergence criteria.

Table 4
Parameters of $\bar{a}_{d,y,t}$ and $\underline{a}_{d,y,t}$.

Coefficient (PB/h)	Time Period(h)					
	1–4	5–8	9–12	13–16	17–20	21–24
$\bar{a}_{1,1,t}$	150	200	200	270	270	180
$\underline{a}_{1,1,t}$	120	150	160	200	180	140
$\bar{a}_{2,1,t}$	120	120	150	220	200	150
$\underline{a}_{2,1,t}$	80	100	100	150	120	100
$\bar{a}_{3,1,t}$	140	180	180	160	160	120
$\underline{a}_{3,1,t}$	100	150	160	120	120	100

Table 2
Parameters of IEWS.

Parameter	Value	Parameter	Value
$c^{\text{e,curt}}$	3.5 ¥/kWh	a_p^{Iwp}	−1.1
$c^{\text{w,curt}}$	8.4 ¥/m ³	b_p^{Iwp}	7448
$V/\bar{V}$	9.7/10.3 kV	n_p^{Iwp}	0.5

Algorithm 3. Procedures of proposed planning framework.**Algorithm 3:** Procedures of proposed planning framework

Initialization:

Initialize once and twice Lagrangian multipliers γ and ρ ;

Set initial integer variable $\mathbf{x}_k^{(0)}$ relaxed to continuous;

Set iteration index $w = 0$.

Repeat:

Step 1: Solve relaxed PC&CG problem:

Solve relaxed PC&CG problem (**Algorithm 1**) under ATC framework;

Obtain initial interactive power/water supply $\mathbf{q}_k^{(w)}$.

Step 2: Solve $\mathbf{q}$ -fixed PC&CG problem:

Fix $\mathbf{q}_k = \mathbf{q}_k^{(w)}$;

Use **Algorithm 1** to optimize integer variable $\mathbf{x}_k^{(w)}$.

Step 3: Solve $\mathbf{x}$ -fixed problem:

Fix $\mathbf{x}_k = \mathbf{x}_k^{(w)}$;

Use **Algorithm 2** to update new $\mathbf{q}_k^{(w+1)}$.

Step 4: Check convergence:

If $\mathbf{x}_k^{(w+1)} = \mathbf{x}_k^{(w)}$, then:

Converged; output $\mathbf{x}_k^{(w)}$ and $\mathbf{q}_k^{(w+1)}$; stop;

Else:

Set $w = w + 1$; go to Step 2.

Until convergence.

The computational complexity of the proposed algorithm is detailed in Appendix B.

It is worth noting that theoretically, due to the non-convex nature of the problem, the outer AOP framework guarantees convergence to a stationary point. However, relying on the global optimality of the inner PC&CG and ATC subproblems, the proposed method effectively approximates the global optimum with a high-quality solution.

5. Case studies

5.1. Parameter settings

In this section, the effectiveness of the proposed method is validated through numerical experiments conducted on a test system. The electric-water system consists of an enhanced IEEE 33-bus distribution network and a 15-node water distribution network, including 32 transmission lines, 14 water pipelines, 6 DGs, 4 WTGs, 1 transformer, 3 water pumps, and a single water source. The IDC system comprises three individual IDC units. The topology of the test system is illustrated in Fig. 3. A 10-year planning horizon is assumed, with an annual urban Multi-energy demands growth rate of 5%. The key parameters of the electric-water

system are listed in Table 2, and those of the IDC system are summarized in Table 3. Additional parameters for both systems can be derived from [7,8,43].

Table 4 presents the DDU coefficients $\bar{\alpha}_{d,y,t}$ and $\underline{\alpha}_{d,y,t}$ for the data load in each IDC area for the first year. We set $\bar{\beta}_{y,t} = \sum_{d \in \Omega^b} \bar{\alpha}_{d,y,t} / 3$ and $\underline{\beta}_{y,t} = \sum_{d \in \Omega^b} \underline{\alpha}_{d,y,t} / 3$. The DDU parameters for line and DG failures, $\Gamma_{i,y}^{\text{ag}}$ and $\Gamma_{i,y}^{\text{ag}}$, are set to 2 and 1, respectively. The remaining DIU factors, including RES output as well as power and water load uncertainties, can be found in [44,45]. To balance computational tractability and seasonal representativeness, each year within the 10-year planning horizon is modeled using 12 typical days, with one representative 24-h profile selected for each month.

5.2. Planning results

This study demonstrates the superiority of the proposed synchronized decentralized planning method through four case studies:

Case I. Centralized planning without considering the differing interests of stakeholders. This scenario can be directly solved using the PC&CG algorithm (**Algorithm 1**).

Case II. Asynchronous planning in which initial values $\gamma_{d,y,t}^{\text{DC-ele}} = \gamma_{d,y,t}^{\text{DC-wat}} = 0.001$ and $\gamma_{d,y,t}^{\text{IEWS-ele}} = \gamma_{d,y,t}^{\text{IEWS-wat}} = 0.2$ are set for each stakeholder, with the IDC taking the lead in planning and dominating the interaction process.

Case III. Asynchronous planning in which initial values $\gamma_{d,y,t}^{\text{DC-ele}} = \gamma_{d,y,t}^{\text{DC-wat}} = 0.2$ and $\gamma_{d,y,t}^{\text{IEWS-ele}} = \gamma_{d,y,t}^{\text{IEWS-wat}} = 0.001$ are set for each stakeholder, but the IEWS leads the planning and dominates the interaction.

Case IV. The synchronized decentralized planning approach proposed in this paper, where initial values $\gamma_{d,y,t}^{\text{DC-ele}} = \gamma_{d,y,t}^{\text{DC-wat}} = \gamma_{d,y,t}^{\text{IEWS-ele}} = \gamma_{d,y,t}^{\text{IEWS-wat}} = 0.001$ are set to ensure equal participation and decision-making power among stakeholders.

Table 5 summarizes the planning results for the four case studies. As shown in Table 5, the total costs over the 10-year planning horizon for Cases I-IV are 156.49 million, 167.02 million, 177.34million, and 160.7

Table 5
Planning outcomes of four Cases.

	Type	IDC	IEWS	Total
Case I	Investment Cost (10 ⁶ ¥)	34.56	12.48	47.04
	Interaction Cost (10 ⁶ ¥)	0	49.83	49.83
	Operational Cost (10 ⁶ ¥)	18.46	39.14	57.6
	Total Cost (10 ⁶ ¥)	53.02	101.45	154.47
Case II	Investment Cost (10 ⁶ ¥)	34.93	17.22	52.15
	Interaction Cost (10 ⁶ ¥)	39.15	12.63	52.38
	Operational Cost (10 ⁶ ¥)	19.96	43.14	63.1
	Total Cost (10 ⁶ ¥)	94.03	72.99	167.02
Case III	Investment Cost (10 ⁶ ¥)	42.39	15.67	58.06
	Interaction Cost (10 ⁶ ¥)	44.96	8.94	53.9
	Operational Cost (10 ⁶ ¥)	26.05	40.34	66.39
	Total Cost (10 ⁶ ¥)	113.4	64.95	178.35
Case IV	Investment Cost (10 ⁶ ¥)	35.27	14.78	50.05
	Interaction Cost (10 ⁶ ¥)	40.53	10.32	50.85
	Operational Cost (10 ⁶ ¥)	19.86	39.94	59.8
	Total Cost (10 ⁶ ¥)	95.66	65.04	160.7

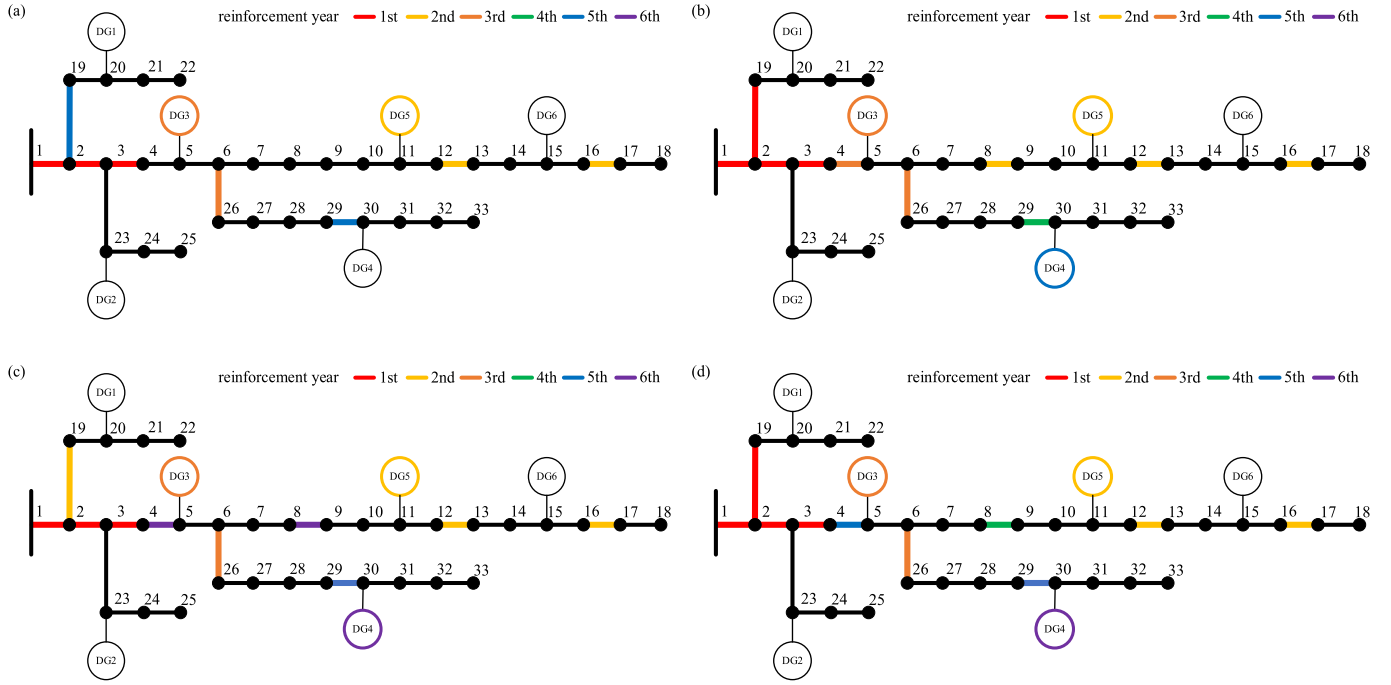


Fig. 4. Planning schemes of IEWS in 10th year for different cases: (a) Case I; (b) Case II; (c) Case III; (d) Case IV.

million, respectively. While the synchronized decentralized planning in Case IV yields slightly higher total costs compared to centralized planning in Case I, it provides an optimal solution under the non-convexity of the synchronized decentralized problem (Only 4.03% deviation from the

centralized method), with guaranteed convergence. This comparison also serves as a numerical verification of the proposed AOP algorithm, demonstrating its capability to find a high-quality solution close to the global optimum found by the centralized approach. Compared to

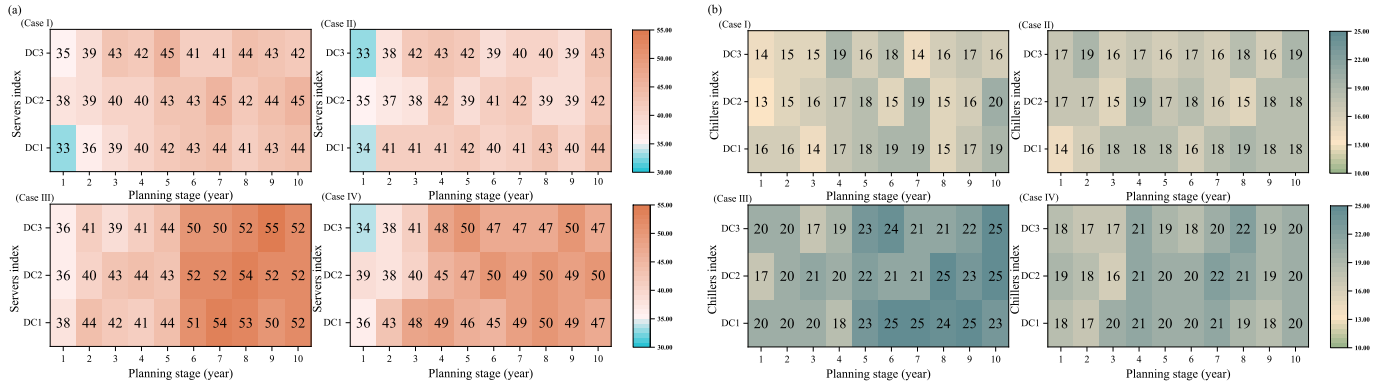


Fig. 5. Planning schemes of IDC in 10th year for different cases: (a) Servers; (b) Chillers.

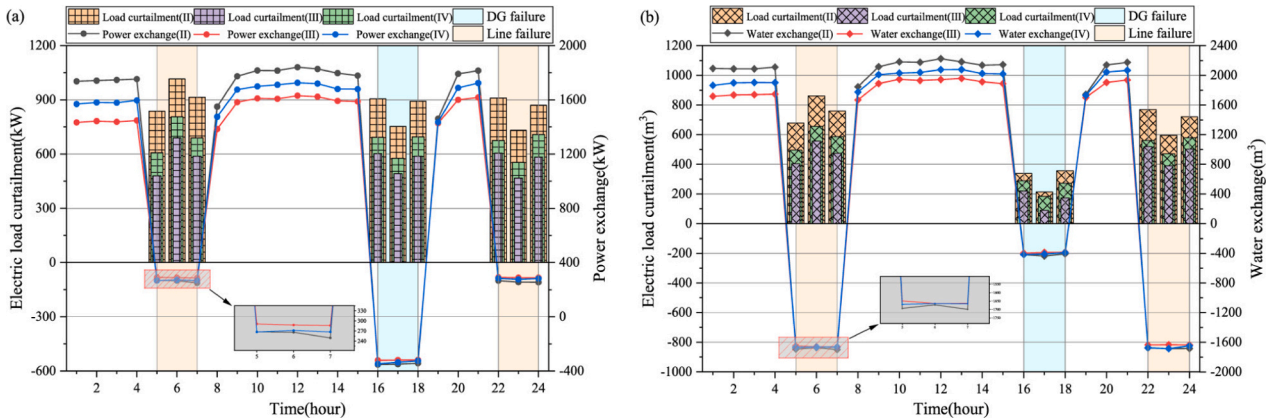


Fig. 6. Interaction and operation outcomes of IDC and IEWS on a representative day in the 10-th year: (a) Power; (b) Water.

Table 6
Planning outcomes of three Cases.

	Type	IDC	IEWS	Total
Case A	Investment Cost (10 ⁶ ¥)	34.17	46.88	81.05
	Interaction Cost (10 ⁶ ¥)	52.46	1.03	53.49
	Operational Cost (10 ⁶ ¥)	14.96	61.22	76.18
	Total Cost (10 ⁶ ¥)	101.59	109.13	210.72
Case B	Investment Cost (10 ⁶ ¥)	35.52	25.87	61.39
	Interaction Cost (10 ⁶ ¥)	47.23	4.88	52.11
	Operational Cost (10 ⁶ ¥)	17.46	48.24	65.7
	Total Cost (10 ⁶ ¥)	100.21	78.99	179.2
Case C	Investment Cost (10 ⁶ ¥)	35.27	14.78	50.05
	Interaction Cost (10 ⁶ ¥)	40.53	10.32	50.85
	Operational Cost (10 ⁶ ¥)	19.86	39.94	59.8
	Total Cost (10 ⁶ ¥)	95.66	65.04	160.7

asynchronous decentralized approaches in Cases II and III, the cost performance of Case IV is closer to that of Case I. In Case II, the IDC incurs lower total costs than in Case IV, while the IEWS experiences higher costs due to IDC's priority in the planning process. A similar trend is observed in Case III, where the IEWS achieves better economic performance due to its leading role in planning, but the overall system cost remains higher than in Case IV. These findings highlight that in asynchronous decentralized planning, prioritizing one stakeholder inevitably compromises the interests of the other, which undermines the coordinated development of the entire system. In contrast, the synchronized decentralized planning approach allows all stakeholders to plan simultaneously, ensuring fairness and balanced participation. It also demonstrates superior overall economic performance compared to asynchronous methods. This underscores the advantage of synchronized decentralized planning in addressing multi-stakeholder decision-making problems.

Fig. 4 (a), (b), (c), and (d) illustrate the IEWS planning results in the 10th year under Cases I–IV, respectively. It can be observed that, regardless of the scenario, network and DG reinforcements are primarily concentrated within the first six years. Additionally, the final planning results in Cases II–IV are identical, indicating that further reinforcements beyond the 6th year yield diminishing returns for the system. In the centralized planning scenario (Case I), the power and water exchanges between the IEWS and IDC are cost-free and highly flexible. As a result, the number of reinforced lines and DG units is the lowest in Case I. In contrast, under decentralized planning, interactive power and water supplies are priced, and in the event of failures, the IEWS acts as a buyer. Compared to Case I, Case II exhibits additional reinforcement on lines 4–5 and 8–9, and DG4, since the IEWS cannot obtain sufficient support

from the IDC. In Case III, the IEWS has planning priority and therefore receives more support from the IDC. Consequently, reinforcements for lines 2–19, 4–5, 8–9, 29–30, and DG4 occur later than in Case II. Similarly, in Case IV, reinforcements of lines 4–5, 8–9, 29–30, and DG4 are also delayed compared to Case II. However, due to the synchronized nature of the planning, the process is more equitable. The IDC provides slightly less support than in Case III, leading to earlier reinforcements of lines 2–19, 4–5, and 8–9 compared to Case III. These results suggest that while the planning order affects the timing of network reinforcements, it has limited impact on the final outcomes over a long-term planning horizon.

Fig. 5 (a) and (b) illustrate the 10-year planning results for servers and chillers in the IDC system, respectively. Similar to the IEWS planning trends, comparable patterns are observed. In Case II, where the planning is led by the IDC, the IDC is not obligated to fully support the IEWS during failure events. As a result, the IDC does not require high flexibility and thus reduces the configuration of servers and chillers. In fact, during the sixth and seventh years, the number of newly added servers and chillers is even lower than in the centralized planning scenario. In Case III, where the IEWS leads the planning process, the IDC invests in the highest number of servers and chillers in order to maximize support for the reliable operation of the IEWS. Compared to Case II, the IDC increases the number of servers by 16.42% and chillers by 26.46%, aiming to shift data loads away from failure periods or affected zones to alternative time slots or more distant IDCs. Case IV falls between Cases II and III in terms of investment levels, showing the closest alignment with the centralized planning configuration. These IDC planning results further confirm that the order of planning influences not only the reinforcement timing but also the infrastructure configuration decisions within the IDC system.

Fig. 6 illustrates the interaction and operational results under the worst-case scenario in the 10th year, with line failures occurring between 5:00–7:00 and 22:00–24:00, and DG failure between 16:00–18:00. These periods are distinguished by different background colors in the figure. Since the centralized optimization yields slightly different results from the planning in this paper for the 10th year, the worst-case scenarios under DDU also vary. Therefore, only the results for Cases II–IV are presented here. In Case II, during failure events, the IDC provides the least support to the IEWS in terms of power and water supply, while also receiving the least load from the IEWS during normal operation. As a result, the IDC's data load distribution is the most even, requiring minimal demand response and being most beneficial to the IDC's operation. Case III shows the opposite trend, with the operation favoring the IEWS. As indicated by the load shedding results, Case III

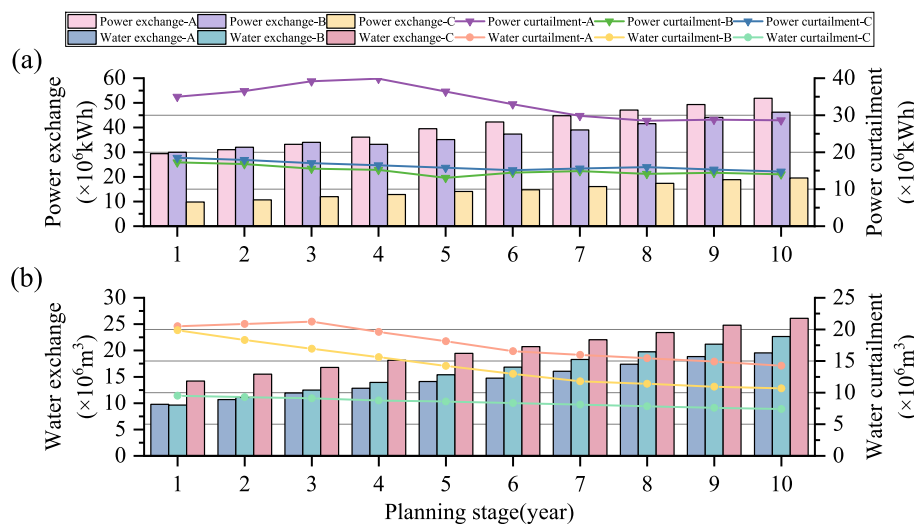


Fig. 7. Support and curtailment outcomes during each planning stage: (a) Power; (b) Water.

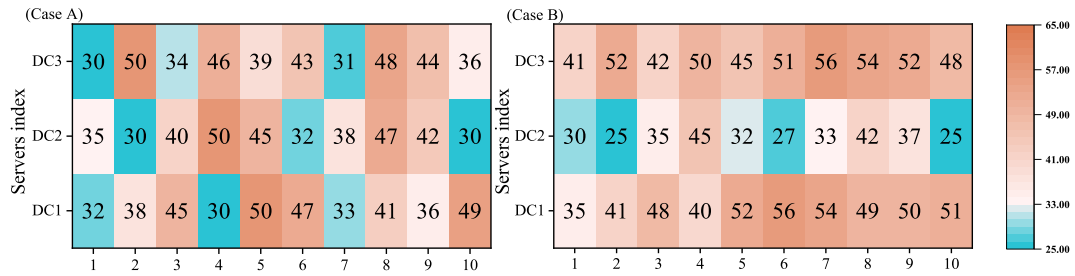


Fig. 8. Number of servers during each planning stage.

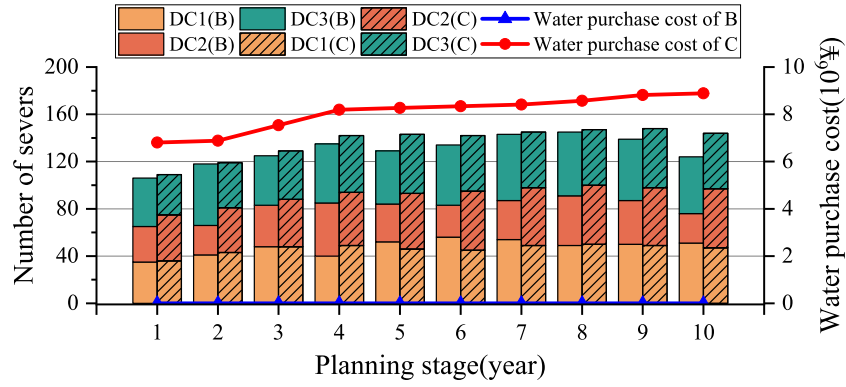


Fig. 9. Planning and operation outcomes of IDC.

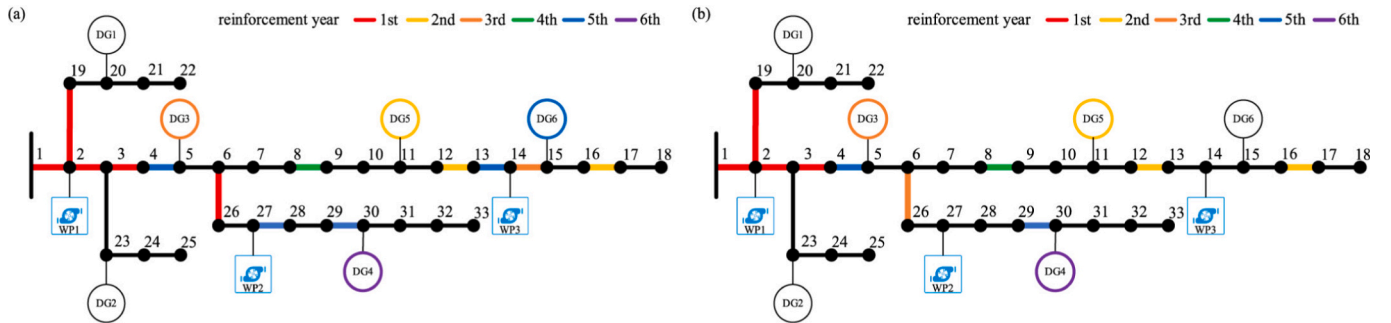


Fig. 10. Planning schemes of IEWS in 10th year for different cases: (a) Case B; (b) Case C.

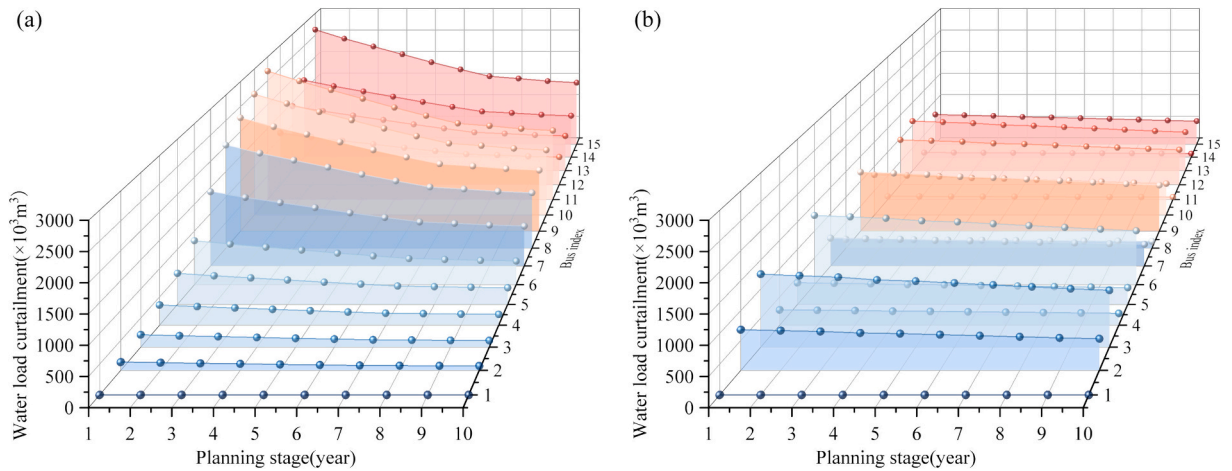


Fig. 11. Water load curtailment at each node over ten years for different cases: (a) Case B; (b) Case C.

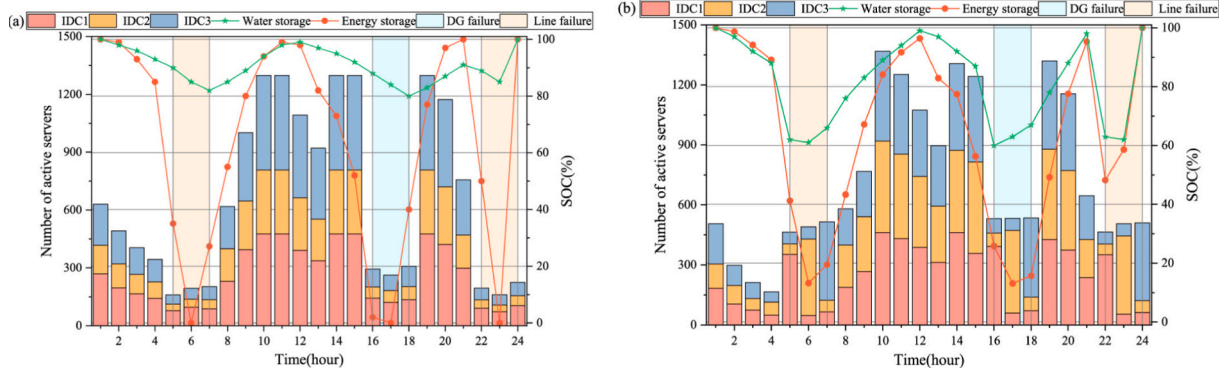


Fig. 12. Number of active servers and status of electric and water energy storage due to a representative day in year 10 for different cases: (a) Case B; (b) Case C.

shows that the IEWS experiences the least load shedding during failure events. **Case IV** strikes a balance between the interests of the IDC and IEWS, with results falling in between those of **Cases II and III**. By comparing Fig. 6 (a) and (b), it can be observed that the IDC tends to provide more water support during line failures, while offering more power support during DG failures. This is because, after ten years of reinforcement, the DGs that are closer to the IDC have already been strengthened, reducing the impact of failing DGs on the water system. In contrast, line failures affect a larger area and are more likely to impact the water pumps, as seen in Fig. 4. Additionally, it is evident that the IDC can provide more water support, and in many cases, the IDC can only reduce electricity consumption rather than provide reverse power. This is because the IDC's water storage capacity is significantly larger than its electrical storage capacity.

5.3. Benefits of considering IDC flexibility support

As a key multifunctional energy hub, the operational scheduling strategy of the IDC not only affects its own operational efficiency but also has a profound impact on the overall reliability and synergy of the IEWS. To validate the collaborative support effect of the IDC on the IEWS in both the electric and water energy dimensions, this study designs and analyzes the following three typical scenarios:

Case A. The IDC functions solely as an energy consumer and does not provide any reverse support to the IEWS, i.e., in the mathematical model, $P_{d,y,t}^{DC} \geq 0, P_{d,y,t}^{DC} \geq 0, f_{d,y,t}^{DC} \geq 0, f_{d,y,t}^{DC} \geq 0$.

Case B. The IDC provides power support but does not contribute to water resource support.

Case C. The IDC provides both power and water resource support in a collaborative manner.

The planning results of the three cases are summarized in Table 6. It can be observed that as the support mechanism evolves from **Case A** to

Table 7

Planning outcomes of three Cases.

	Type	IDC	IEWS	Total
#1	Investment Cost (10^6¥)	32.09	12.42	44.51
	Interaction Cost (10^6¥)	41.11	12.91	54.02
	Operational Cost (10^6¥)	19.15	63.12	82.27
	Total Cost (10^6¥)	92.35	88.45	180.8
#2	Investment Cost (10^6¥)	32.16	14.62	46.78
	Interaction Cost (10^6¥)	41.13	9.98	51.11
	Operational Cost (10^6¥)	19.16	41.25	60.41
	Total Cost (10^6¥)	92.45	65.85	158.3
#3	Investment Cost (10^6¥)	35.27	14.78	50.05
	Interaction Cost (10^6¥)	40.53	10.32	50.85
	Operational Cost (10^6¥)	19.86	39.94	59.8
	Total Cost (10^6¥)	95.66	65.04	160.7

Case C, both power and water load shedding in the IEWS gradually decrease, while the corresponding interactive energy quantities increase. The load shedding rate $\varphi^e = (\sum_{y \in \Omega^Y} \sum_{i \in \Omega^I} \sum_{t \in \Omega^T} P_{i,y,t}^{e,curt}) / \sum_{y \in \Omega^Y} \sum_{i \in \Omega^I} \sum_{t \in \Omega^T} P_{i,y,t}^L$, $\varphi^w = (\sum_{y \in \Omega^Y} \sum_{i \in \Omega^W} \sum_{t \in \Omega^T} f_{i,y,t}^{w,curt}) / \sum_{y \in \Omega^Y} \sum_{i \in \Omega^W} \sum_{t \in \Omega^T} f_{i,y,t}^L$ is defined to quantify this effect. The power load shedding rates for **Cases A, B, and C** are 4.31%, 1.19%, and 1.29%, respectively, while the water load shedding rates are 6.33%, 4.27%, and 2.15%, respectively.

First, the comparison between **Cases A and B** confirm the benefit of enabling the IDC to provide power support. The annual power and water exchange, as well as load curtailment under all three scenarios, are shown in Fig. 7. Compared to **Case A**, **Case B** achieves a reduction of 185.84 MWh in power load shedding. This improvement arises from the alignment between the IDC investor's profit-driven behavior and the IEWS's demand for electricity during failure periods. Taking server deployment as an example, Fig. 8 illustrates the server configuration results under different scenarios. In **Case B**, the introduction of electric support leads to higher power demand at Node 26, which increases during failure events. To address this, RES investors deploy more servers

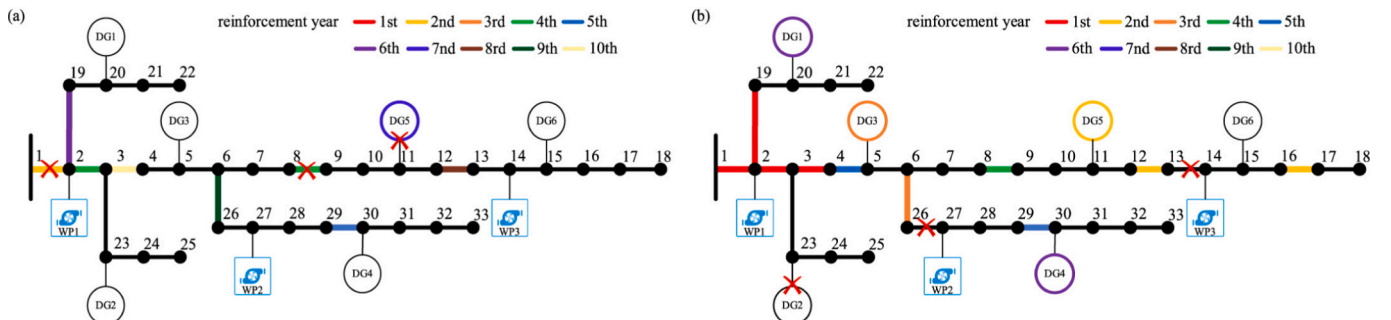


Fig. 13. Planning schemes of IEWS in 10th year for different cases: (a) #1; (b) #2.

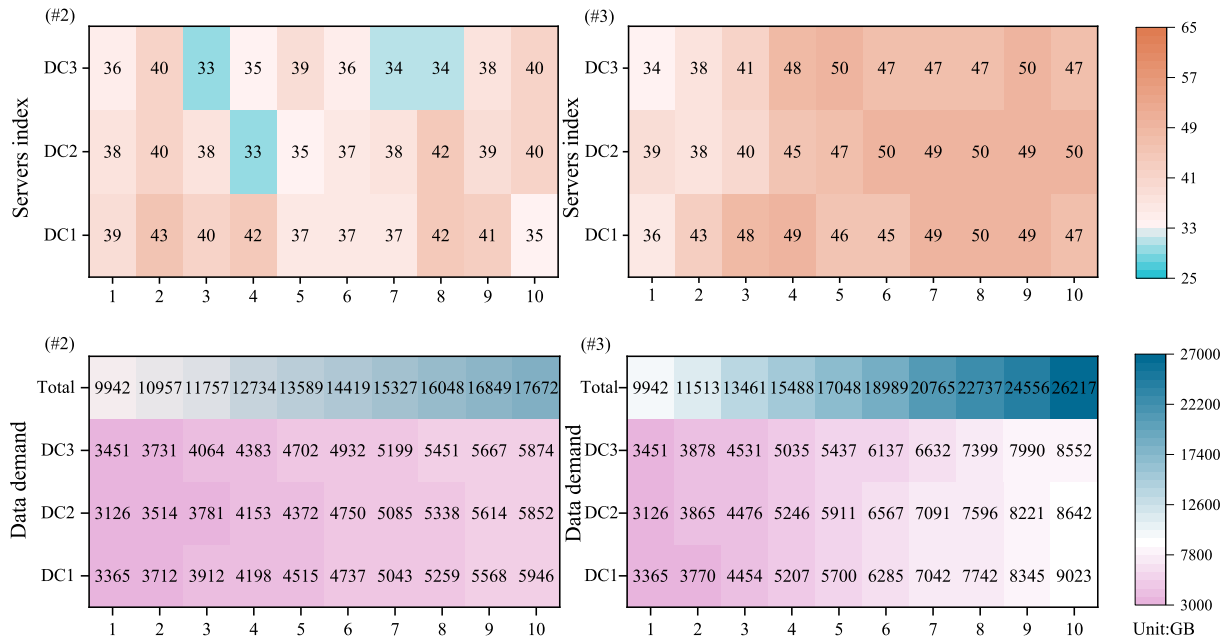


Fig. 14. Planning schemes and data demand of IDC in 10th year for different cases.

in IDC1 and IDC3-located farther from Node 26-compared to Case A. This strategy allows IDC2, which is closer to Node 26, to offload data during failures and use its local energy storage to supply electric support. This reallocation not only maximizes the IDC's profit but also significantly mitigates power shedding in the IEWS.

Moreover, the comparison between Cases B and C highlights the importance of water support for both IDC and IEWS investments. As shown in Fig. 9, starting from Year 1, the IEWS invests 6.81 million to purchase water from the IDC in order to reduce water load shedding during failure periods. In response, the IDC installs additional servers to enhance its demand response capability. Due to the increased water support provided by the IDC, the reinforcement of lines near water pumps in Case C is delayed or even avoided, as illustrated in Fig. 10.

Fig. 11 demonstrates that the introduction of water support significantly reduces the water load shedding in the IEWS. Additionally, as shown in Fig. 7, although the IDC reallocates part of its resources from electric support to water support, the overall impact on electric performance is minimal. This is because direct water support is far more efficient at reducing water shortages than indirect support via the power grid to power water pumps. As a result, only a small portion of the IDC's energy storage needs to be sacrificed, and the power load shedding in Case C is only marginally higher than in Case B. Finally, Fig. 12 presents the IDC's operational behavior during a failure event. It reveals that the IDC primarily relies on temporal shifting to provide electric support, while spatial shifting is the dominant strategy for water support. This is because the water tanks in IDCs typically have significantly higher capacity compared to electric energy storage. Consequently, when offering water support, the IDC can prioritize spatial shifting, which incurs no cost. In contrast, due to limited storage capacity, the IDC cannot rely solely on spatial shifting for electricity and must instead depend more heavily on temporal shifting to maintain its own operations.

A holistic comparison of the three cases reveals a non linear relationship between support diversity and subsystem reliability which exposes critical resource limitations. Case A establishes the baseline vulnerability where passive load consumption leads to maximum shedding. The transition to Case B yields the most significant marginal benefit reducing power shedding by 72.39% by addressing the root cause of failures. However, the transition to Case C reveals a counter intuitive phenomenon where electrical load shedding actually increases slightly from 1.19% to 1.29% compared to Case B. This observation

Table 8

Economic benefits assessment of IDC.

Index	Total cost (10 ⁶ ¥)	Investment cost (10 ⁶ ¥)	Operational cost (10 ⁶ ¥)	V_{DIE} (10 ⁶ ¥)	$\bar{V}_{DIE}$
#2	-115.29	32.16	-147.45	-	-
#2*	-120.58	32.16	-152.74	-	-
#3	-145.96	35.27	-181.23	25.38	21.05%

highlights the physical constraints of the IDC's finite energy storage. To provide water support, the IDC must divert a portion of its battery power to drive internal water pumps, thereby reducing the available energy for grid back-feeding. Despite this slight degradation in electrical performance Case C achieves the highest global reliability by reducing water shedding by nearly 50% compared to Case B. This demonstrates that achieving optimal cross sectoral resilience requires a strategic compromise sacrificing single domain electrical optimality to secure overall system stability.

It is worth noting that a scenario where the IDC provides only water support without electrical feedback was also evaluated. In such a scenario the electrical operational state is identical to the baseline no support strategy defined as Case A. As shown by the results of Case A in Fig. 7a and Table 6 without electrical feedback from the IDC the power outages that disable municipal water pumps cannot be mitigated. Consequently, the electrical load shedding remains at the high level of 4.31% which is identical to Case A. Furthermore, the water network recovery is limited solely to the hydraulic capacity of the IDC tanks as it lacks the systemic recovery driven by restarted pumps. Consequently, while water-only support offers marginal benefits over the baseline it is strictly inferior to the strategies involving power support in Case B and Case C. Thus, it is not presented as a separate curve in the comparative results.

5.4. Comparison of uncertainty modeling mechanisms

To comprehensively evaluate the robustness and effectiveness of the proposed planning method under different uncertainty modeling frameworks, three modeling scenarios are designed for comparative analysis:

#1: A traditional robust optimization model that treats all un-

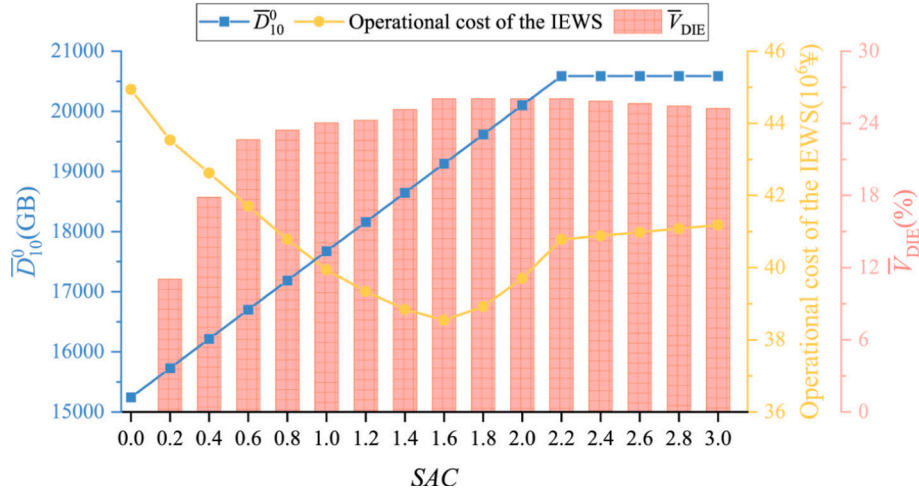


Fig. 15. Sensitivity of demand inducing effect.

certainties as exogenous factors. Mathematically, this is implemented by setting $\bar{\alpha}_{d,y,t} = 0, \underline{\alpha}_{d,y,t} = 0, \bar{\beta}_{y,t} = 0$ and $\underline{\beta}_{y,t} = 0$, and removing constraints (68) and (70).

#2: An enhanced model that incorporates DDU to capture uncertainties associated with distribution system failures.

#3: The full modeling framework proposed in this study, which utilizes DDU to characterize both distribution system failure scenarios and IDC load fluctuations.

Fig. 13 compares the IEWS planning results under scenarios #1 and #2. In scenario #1, reinforcement actions are triggered only when load growth over time causes line congestion or generation capacity becomes insufficient. As a result, the first reinforcement occurs in Year 2, with most upgrades concentrated in the latter half of the planning horizon. In contrast, scenario #2 reflects the influence of DDU modeling of distribution system failures, leading to concentrated reinforcements in the first five years. Notably, the worst-case failure scenarios in #1, such as line failures on 1–2 and 8–9, and DG5, are preemptively addressed in scenario #2 through early reinforcements. To further evaluate the economic implications of IEWS planning, we analyzed three key indicators: IEWS investment cost, interaction cost, and the penalty cost due to unmet load during operation. As shown in Table 7, the total operational cost in #2 is only 9.98 million, supporting the economic feasibility of proactive investment in the IEWS since lower operating costs translate into greater profitability over the planning period. Compared to #1,

scenario #2 achieves a 32.62% reduction in the combined interaction and operational costs, while the investment cost increases by only 17.71%. This result clearly demonstrates that incorporating DDU to model uncertainty in the IEWS planning framework significantly enhances economic efficiency.

Fig. 14 presents the IDC planning results for scenarios #2 and #3, along with the average satisfied data load demand at the regional level $\bar{D}_{d,y}^0 = \sum_{t \in \Omega^T} D_{d,y,t}^0 / \text{Time}$. In scenario #2, a total of 1138 new servers are deployed, while in scenario #3, the number increases to 1368. The corresponding system-wide average satisfied data loads $\bar{D}_y^0 = \sum_{d \in \Omega^D} \bar{D}_{d,y}^0$ is also shown in Fig. 14. Moreover, in scenario #3, the trend of server deployment accelerates over the planning horizon, with a significantly larger increment in the later years. Specifically, in the 10th year, scenario #3 achieves an additional 8545 GB of $\bar{D}_y^0$ compared to scenario #2, clearly illustrating the demand inducing effect. This phenomenon can be interpreted as follows: server deployment stimulates the growth of data load demand, which in turn drives further investment in IDC server capacity. This mutual reinforcement eventually leads to a new equilibrium between supply and demand. Using the data load pricing model from [46] to evaluate IDC revenue, we compare the total cost, investment cost, and operational cost of IDC planning under scenarios #2 and #3, as shown in Table 7. In scenario #3, the total cost amounts to −145.96 million, indicating the economic viability of IDC investments-

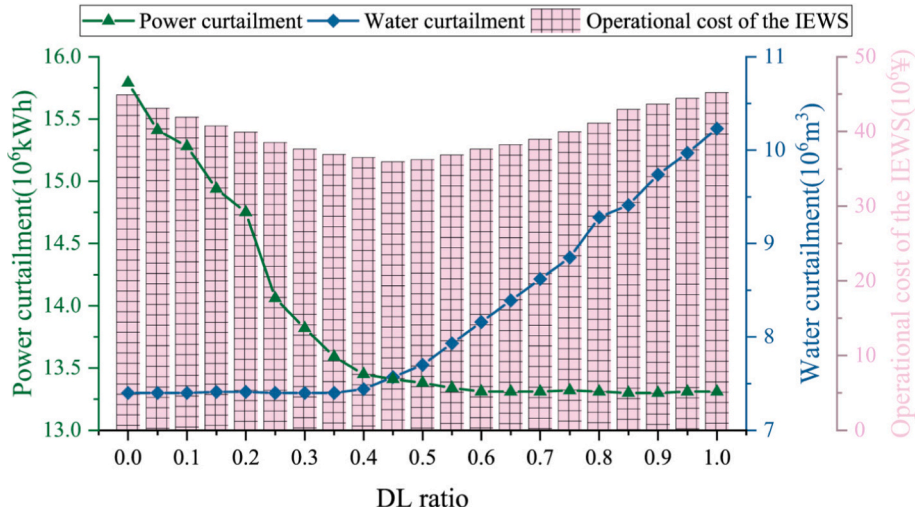


Fig. 16. Sensitivity of DL ratio.

since a negative net cost implies that positive profits can be achieved over the planning horizon. Compared to #2, scenario #3 reduces operational costs by 22.91%, while investment costs increased by only 9.67%. This demonstrates that incorporating DDU into IDC modeling enhances flexibility and economic performance.

In the three-level optimization model, the intermediate-level problem tends to seek the lowest possible data load demand scenario, thereby limiting the IDC's demand response capability. However, with the introduction of the Data-Driven Uncertainty (DDU) set, the "worst-case" data load demand increases with the number of deployed servers, effectively enhancing the investment return. To further analyze this effect, we introduce scenario #2*, which represents the expected economic performance of scenario #2 when the demand inducing effect is taken into account. Specifically, the investment strategy from #2 is held fixed, and the DDU model is re-solved under this constraint. A comparison between #2 and #2* reveals that accounting for induced demand results in a cost reduction of 5.29 million, indicating that planning based solely on a DIU set may underestimate the true economic benefits. Consequently, such underestimation may lead to the rejection of investment projects that would, in reality, yield positive returns. To quantitatively assess this benefit, we introduce two evaluation metrics defined in [47]:

The Value of Demand inducing effect:

$$V_{DIE} = \varphi_{\#2^*} - \varphi_{\#3} \quad (113)$$

Relative Value of the Demand inducing effect:

$$\bar{V}_{DIE} = \frac{\varphi_{\#2^*} - \varphi_{\#3}}{|\varphi_{\#2^*}|} \times 100\% \quad (114)$$

Let $\varphi_{\#2^*}$ and $\varphi_{\#3}$ denote the total IDC costs under scenarios #2* and #3, respectively. By definition, and considering that the result of #2* is also a feasible solution under scenario #3, we can conclude that: $\varphi_{\#2^*} \geq \varphi_{\#3}$. As shown in Table 8, compared to #2*, scenario #3 achieves a cost saving of 25.38 million in terms of V_{DIE} , and an improvement of 21.05% in the relative value $\bar{V}_{DIE}$.

A joint analysis of the three scenarios illustrates the progressive value of internalizing the coupling mechanisms between decisions and uncertainties. #1 treats uncertainties as static exogenous inputs resulting in a conservative planning scheme that misses the potential for demand stimulation. #2 partially addresses this by incorporating failure related DDU which optimizes the timing of reliability reinforcements. Scenario #3 integrates the full DDU framework capturing the demand inducing effect where investment actively drives load growth and revenue. Consequently, #3 achieves the best economic performance by transforming reliability investments into value generating assets. However, it is critical to acknowledge the limitations of this DDU framework.

The economic superiority of #3 relies heavily on the accurate estimation of the demand induction coefficients. If these parameters are over-estimated due to insufficient historical data or market saturation, the model may face the risk of over-investment. Therefore, the proposed method requires high quality data support and sensitivity checking to ensure the validity of the planning decisions.

5.5. Analysis of influencing factors

5.5.1. Impact of demand-induced effect

We further conducted a sensitivity analysis on the demand inducing effect based on the DDU parameters. Let $\bar{\alpha}_{d,y,t} = SAC \bar{\alpha}_{d,y,t}^0$ and $\underline{\alpha}_{d,y,t} = SAC \underline{\alpha}_{d,y,t}^0$, where $SAC \in [0, 3]$, $\bar{\alpha}_{d,y,t}^0$ and $\underline{\alpha}_{d,y,t}^0$ represent the baseline values as defined in Section 4.1 Table 4. Fig. 15 illustrates the 10th-year data load demand $\bar{D}_{10}^0$, the operational cost of the IEWS, and $\bar{V}_{DIE}$ under different SAC values. As SAC increases from 0 to 2.2, $\bar{D}_{10}^0$ gradually increases, reflecting an improvement in the system's data load processing capacity. In this case, the marginal effect of $\bar{D}_{10}^0$ remains constant, increasing by 486 GB for every 0.2 increase in SAC. When $SAC \geq 2.2$, $\bar{D}_{10}^0$ stabilizes at 20,588 GB, primarily due to the limitations of the system's power and water supply capacity at the nodes where the IDC is located. On the other hand, as SAC increases from 0 to 1.6, the operational cost of the IEWS shows a negative correlation with SAC, decreasing from 44.94 million to 38.54 million. When SAC increases from 0 to 1, the operational cost decreases by 5 million, and when SAC increases from 1 to 1.6, the reduction in operational cost is only 1.4 million. This suggests that as SAC increases and the IDC's demand response capacity improves, the IEWS indeed benefits from higher revenues, but the gains diminish over time. Additionally, when SAC increases from 1.6 to 2.2, the IEWS operating cost begins to rise, due to the excessive data load leading to a decline in the IDC's demand response capacity. At $SAC \geq 2.2$, the IDC nearly loses its demand response capability and can no longer provide reverse support to the IEWS, resulting in no further increase in the operational cost of the IEWS. The variation in $\bar{V}_{DIE}$ shows a growth trend as SAC increases from 0 to 1.6 (increasing from 0 to 26.05%), after which it stabilizes, with $\bar{V}_{DIE}$ decreasing from 26.05% to 25.25% when $SAC \geq 2.2$. Notably, this decrease begins only after SAC stabilizes and is still positive. From this analysis, we can conclude that the contribution of the demand inducing effect to $\bar{V}_{DIE}$ is positive. However, as other system limitations become bottlenecks, this effect gradually weakens.

5.5.2. Impact of data load proportion

When the IDC provides support to the IEWS, the proportion of

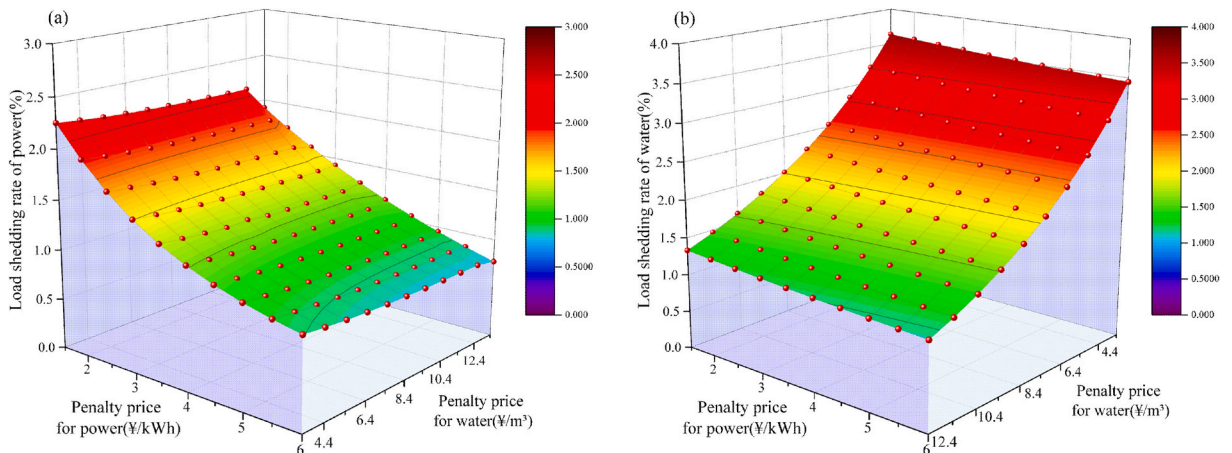


Fig. 17. Sensitivity analysis of load shedding rates with respect to penalty coefficients (a) $c^{e,curt}$; (b) $c^{w,curt}$.

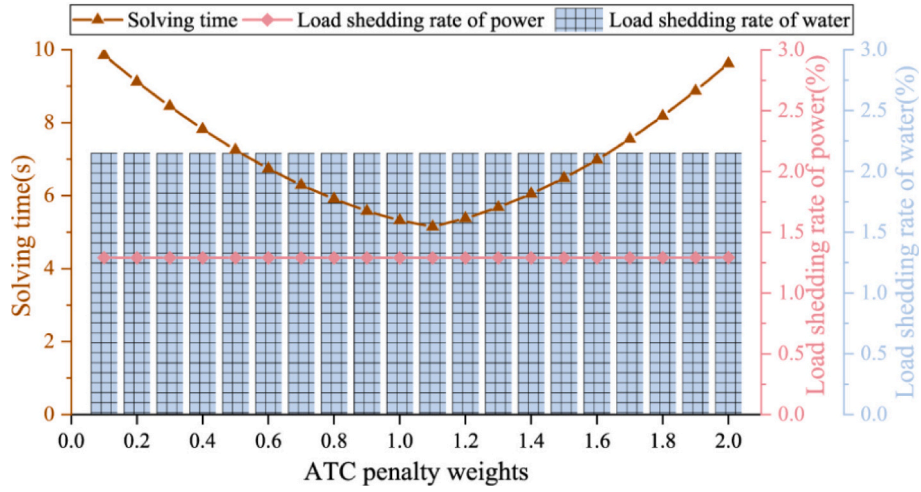


Fig. 18. Sensitivity of solving efficiency and reliability to ATC penalty weights.

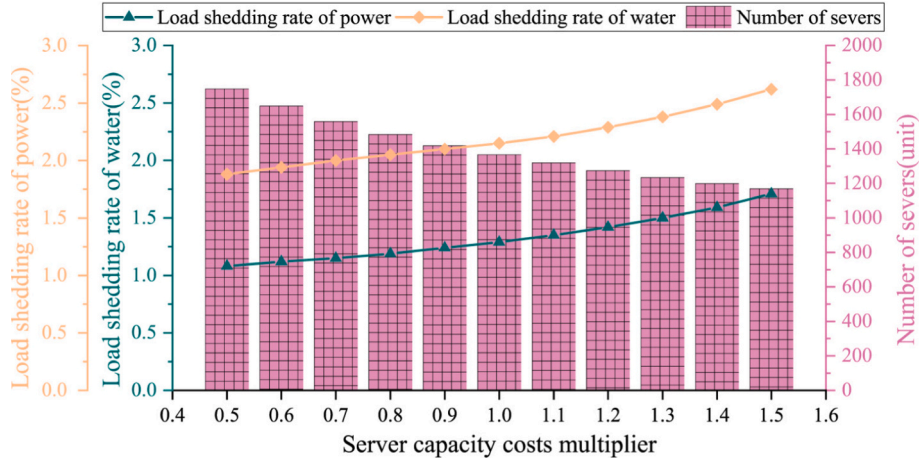


Fig. 19. Sensitivity of number of servers and reliability to server capacity costs.

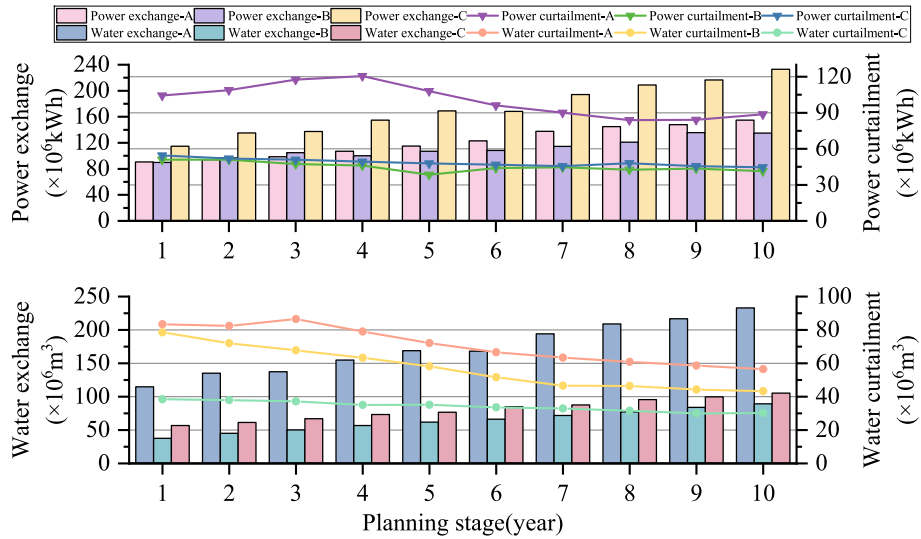


Fig. 20. Support and curtailment outcomes during each planning stage: (a) Power; (b) Water.

different data load types significantly impacts the system's stability and cost. Let the proportion of DL χ^{DL} in the total load be in the range of 0 to 1. Fig. 16 depicts the IEWS operational cost, the reduction in power load, and the reduction in water load for the 10th year under different DL

proportions. As χ^{DL} increases from 0 to 0.3, the power load shedding decreases rapidly, from 15.79 to 13.82 million kWh. When χ^{DL} increases from 0.3 to 0.6, the power load shedding only decreases by 0.51 million kWh. After this point, the power load shedding remains almost

unchanged. The trend in water load shedding is the opposite of power load shedding. As χ^{DL} increases from 0 to 0.35, the reduction in water load remains almost constant. However, when χ^{DL} exceeds 0.35, the reduction in water load increases from 7.4 to 10.23 million m^3 . This also supports the conclusion presented in Section 4.3: the IDC primarily uses its temporal shifting capability to provide power support to the IEWS, while its spatial shifting capacity is used for water support. From the IEWS operational cost curve, it can be observed that when $\chi^{\text{DL}} = 0.45$, the operational cost reaches its minimum, denoted as 35.97 million.

5.5.3. Robustness verification on model parameters

This subsection validates the robustness of the proposed SDRP framework through sensitivity analyses on penalty coefficients, algorithmic settings, and investment costs.

(1) Robustness to Penalty Coefficients.

We examined the impact of varying electricity $c^{\text{e,curt}}$ and water $c^{\text{w,curt}}$ penalty coefficients within ranges of 1.5 to 6.0 ¥/kWh and 4.4 to 13.4 ¥/m^3 , respectively. Fig. 17(a) and Fig. 17(b) illustrate the results. The IEWS demonstrates strong robustness as the electricity load shedding rate only rises to 2.25% even when $c^{\text{e,curt}}$ drops to 1.5 ¥/kWh . This confirms that reliability improvements stem from the intrinsic structural optimization of the SDRP framework and the DDU mechanism rather than prohibitive penalties. Additionally, a positive cross-coupling effect exists where increasing $c^{\text{w,curt}}$ reduces electricity load shedding from 1.34% to 1.25% at the baseline. Similarly, the water load shedding rate in Fig. 17(b) remains controlled at 3.6% under the lowest penalty of 4.4 ¥/m^3 . Increasing $c^{\text{e,curt}}$ also indirectly enhances water reliability by lowering the shedding rate from 2.25% to 2.06%. This synergy validates that a robust power grid effectively safeguards critical water pumps against supply interruptions.

(2) Robustness to Algorithmic Settings.

To demonstrate the stability of the distributed solution algorithm, we performed a sensitivity analysis on the initial ATC penalty weights ρ . Based on the theoretical range, we varied ρ from 0.1 to 2 times the baseline value (set as $\rho = 1$), with a step size of 0.1. The impacts on solving efficiency and solution quality are plotted in Fig. 18.

As illustrated in Fig. 18, the choice of ρ primarily influences the computational efficiency rather than the physical planning outcomes. The solving time exhibits a characteristic U-shaped trend, achieving the highest efficiency (5.15 s) at the baseline setting ($\rho = 1$). Deviating from this optimal point—either by reducing ρ to 0.1 or increasing it to 2—increases the solving time to nearly 10 s due to slower convergence or oscillation. Crucially, however, the electricity and water load shedding rates remain strictly stable at 1.29% and 2.15%, respectively, across the entire range. This verifies that the algorithm consistently converges to the same global optimum regardless of the initialization of penalty weights, ensuring the reliability of the planning results.

We further investigate the impact of server capacity investment costs on the optimal system configuration and reliability. Since the server capacity is a decision variable in the proposed planning-operation model, its unit cost directly affects the optimal number of deployed servers. As shown in Fig. 19, we vary the server capacity cost from 50% to 150% of the benchmark value.

A clear trade-off between economic investment and system reliability is observed. When the capacity cost is low (e.g., 0.5 times), the planner tends to deploy more servers (increasing from 1368 to 1750) to utilize their flexibility for renewable integration, thereby reducing the load shedding rates of power and water systems to 1.08% and 1.88%, respectively. Conversely, as the cost increases to 1.5 times, the optimal server quantity decreases to 1170 to curb investment expenses. Consequently, the reduced flexibility resources lead to a moderate increase in load shedding rates (Power: 1.71%, Water: 2.62%). However, even under high cost scenarios, the load shedding rates remain within an acceptable range.

Table 9

Solving accuracy and efficiency of 3 models.

	Model-0	Model-1	Model-2
φ^{de}	/	4.62%	3.88%
Solving time	624.6 s	36.2 s	59.3 s

5.6. Planning results in the larger-scale system

To verify the effectiveness of the proposed model and algorithm, this section presents simulation results based on a large-scale integrated system. The data center subsystem consists of eight IDCs and twelve information links, while the IEWS subsystem is composed of the IEEE 118-bus power network and a 45-node water network. Detailed parameters are provided in [5,48,49]. Fig. 20 illustrates the annual electricity–water load exchanges and curtailments under the three cases defined in Section 5.3. Similar results can also be observed in larger systems. Through the joint planning of the IDC and IEWS subsystems, Case C achieves a significant reduction in both power and water load curtailments. Specifically, the power load curtailment rates for Cases A, B, and C are 4.79%, 1.08%, and 1.11%, respectively, while the corresponding water load curtailment rates are 5.96%, 4.12%, and 2.08%.

In addition, the accuracy and computational efficiency of the linearized model described in Section 3.2 are evaluated on the large-scale system. The linearized formulation is denoted as Model-1, which represents the method proposed in this study. Eqs. (58) and (61) can further be approximated using first-order Taylor expansion, forming Model-2 [50]. For comparison, a nonconvex formulation, denoted as Model-0, is also considered and directly solved by a commercial nonconvex solver. Using the results obtained from the derived reference model as a benchmark, the following metrics are defined to quantify the solution accuracy of different models:

$$\varphi^{\text{de}} = \max_{y \in \Omega^Y, t \in \Omega^T} \left\{ \varphi_{y,t}^{\text{de-1}}, \varphi_{y,t}^{\text{de-2}}, \varphi_{y,t}^{\text{de-3}}, \varphi_{y,t}^{\text{de-4}}, \varphi_{y,t}^{\text{de-5}} \right\} \times 100\% \quad (115)$$

$\varphi_{y,t}^{\text{de-1}}, \varphi_{y,t}^{\text{de-2}}, \varphi_{y,t}^{\text{de-3}}, \varphi_{y,t}^{\text{de-4}}$ and $\varphi_{y,t}^{\text{de-5}}$ can be calculated by (116)–(120):

$$\varphi_{y,t}^{\text{de-1}} = \max_{(ij) \in \Omega^I} \left\{ \frac{|P_{ij,y,t} - P_{ij,y,t}^0|}{|P_{ij,y,t}^0|} \right\}, \forall y \in \Omega^Y, \forall t \in \Omega^T \quad (116)$$

$$\varphi_{y,t}^{\text{de-2}} = \max_{(ij) \in \Omega^I} \left\{ \frac{|Q_{ij,y,t} - Q_{ij,y,t}^0|}{|Q_{ij,y,t}^0|} \right\}, \forall y \in \Omega^Y, \forall t \in \Omega^T \quad (117)$$

$$\varphi_{y,t}^{\text{de-3}} = \max_{g \in \Omega^G} \left\{ \frac{|P_{g,y,t}^{\text{gen}} - P_{g,y,t}^{\text{gen0}}|}{|P_{g,y,t}^{\text{gen0}}|} \right\}, \forall y \in \Omega^Y, \forall t \in \Omega^T \quad (118)$$

$$\varphi_{y,t}^{\text{de-4}} = \max_{g \in \Omega^G} \left\{ \frac{|Q_{g,y,t}^{\text{gen}} - Q_{g,y,t}^{\text{gen0}}|}{|Q_{g,y,t}^{\text{gen0}}|} \right\}, \forall y \in \Omega^Y, \forall t \in \Omega^T \quad (119)$$

$$\varphi_{y,t}^{\text{de-5}} = \max_{p \in \Omega^P} \left\{ \frac{|f_{p,y,t}^{\text{wp}} - f_{p,y,t}^{\text{wp0}}|}{|f_{p,y,t}^{\text{wp0}}|} \right\}, \forall y \in \Omega^Y, \forall t \in \Omega^T \quad (120)$$

$P_{ij,y,t}^0, Q_{ij,y,t}^0, P_{g,y,t}^{\text{gen0}}, Q_{g,y,t}^{\text{gen0}}$ and $f_{p,y,t}^{\text{wp0}}$ in the equations represent the solutions obtained from Model-0.

Table 9 presents the errors and computational times of the two models. It can be observed that Model-2 yields smaller errors than Model-1. However, since Model-2 involves two sequential solution procedures, its computational speed is slower than that of Model-1. The error difference between Model-1 and Model-3 is 0.74%. In practical applications, decision-makers should flexibly select among these three models to achieve an appropriate trade-off between solution accuracy

and computational efficiency.

6. Conclusion

This study proposes an innovative SDRP framework to optimize the collaborative operation of IDC and IEWS, demonstrating substantial improvements in cost-efficiency, system reliability, and uncertainty handling capabilities. First, the synchronized decentralized planning mechanism effectively coordinates the decision-making of multiple stakeholders while preserving the modeling independence of each subsystem. Compared to centralized solution methods, the proposed distributed solution framework exhibits only a 4.03% deviation in solution quality, while successfully optimizing the costs of IDC and IEWS to 95.66 million and 65.04 million, respectively. Second, the dual-support reliability strategy, which comprises emergency power supply and water tank storage, significantly enhances system performance during grid failure events. Specifically, it reduces electric and water load shedding rates from 4.31% and 6.33% to 1.29% and 2.15%, mitigating the risk of cascading failures across coupled infrastructures. Third, the integration of DDU modeling enables a unified representation of load fluctuations and network faults. This enhances the adaptability and robustness of the scheduling framework, resulting in a 22.91% reduction in IEWS operational costs and a 21.05% improvement in the IDC's data handling capacity. Sensitivity analyses further confirm the robustness and stability of the proposed method. The system achieves optimal performance when the demand induction coefficient is set to 1.6 and the DL ratio reaches 45%, demonstrating its broad applicability under varying operating conditions. In summary, this research provides both theoretical innovation and practical evidence for enhancing collaborative planning among multi-stakeholder infrastructures, and offers a

robust technical solution for improving the reliability of smart city critical systems under uncertainty. The proposed framework can be further extended to distributed robust optimization for resilient decision-making under extreme events, offering practical applicability to the planning and operation of integrated power–water–data infrastructures in smart and disaster-prone cities. Future work could explore the integration of additional water-side uncertainties, such as cooling water evaporation and pumping dynamics, to further enhance the robustness and applicability of the proposed framework in real-world, multi-resource systems.

CRedit authorship contribution statement

Chen Liang: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Bo Zeng:** Writing – review & editing. **Yueyi Lei:** Writing – review & editing. **Zhu Wen:** Writing – review & editing. **Jiayi Zhang:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by the National Natural Science Foundation of China (52177082) and the Beijing Nova Program (20220484007).

Appendix A. We consider the following primal SOCP with multiple SOC constraints

$$\min_{y_k \in Y_k(x_k, q_k, \zeta_k)} d_k^T y_k$$

$$Y_k(x_k, q_k, \zeta_k, \varpi_k) = \{y_k \in \mathbb{R}_+^{n_k} \mid \|M_{ck}y_k + m_{ck1}\|_2 \leq m_{ck2} + n_{ck}^T y_k - o_{ck}^T x_k - p_{ck}^T q_k - q_{ck}^T \zeta_k - r_{ck}^T \varpi_k : u_{ck}\},$$

$$k \in \{DC, IEWS\}, \forall c \in \Omega^C$$
(1)

Each SOC constraint can be written as the conic inclusion:

$$\begin{bmatrix} m_{ck2} + n_{ck}^T y_k - o_{ck}^T x_k - p_{ck}^T q_k - q_{ck}^T \zeta_k - r_{ck}^T \varpi_k \\ M_{ck}y_k + m_{ck1} \end{bmatrix} \in \mathcal{C}^{(m_c+1)}$$

where $\mathcal{C}^{(m_c+1)} := \{(a_1, a_2) \in \mathbb{R} \times \mathbb{R}^{m_c} : a_1 \geq \|a_2\|_2\}$

(2)

Define the affine mapping for each constraint $c \in \Omega^C$.

$$F_c(y_k) := \begin{bmatrix} m_{ck2} + n_{ck}^T y_k - o_{ck}^T x_k - p_{ck}^T q_k - q_{ck}^T \zeta_k - r_{ck}^T \varpi_k \\ M_{ck}y_k + m_{ck1} \end{bmatrix} = K_c y_k + f$$
(3)

$$K_c = \begin{bmatrix} n_{ck}^T \\ M_{ck} \end{bmatrix} \in \mathbb{R}^{(m_c+1) \times n}, f_c = \begin{bmatrix} m_{ck2} - o_{ck}^T x_k - p_{ck}^T q_k - q_{ck}^T \zeta_k - r_{ck}^T \varpi_k \\ m_{ck1} \end{bmatrix} \in \mathbb{R}^{m_c+1}$$
(4)

Introduce dual variables $u_{ck} \in \mathcal{C}^{*(m_c+1)}$ for each SOC constraint. Since the second-order cone is self-dual, we have $\mathcal{C}^{*(m_c+1)} = \mathcal{C}^{(m_c+1)}$. The Lagrangian is:

$$\mathcal{L}(y_k, u_{ck}) = d_k^T y_k + \sum_{c \in \Omega^C} u_{ck}^T (K_c y_k + f_c)$$
(5)

Taking the infimum over x :

$$g(u_{ck}) = \inf_{y_k} \mathcal{L}(y_k, u_{ck}) = \begin{cases} \sum_{c \in \Omega^C} u_{ck}^T f_c, & \text{if } d_k + \sum_{c \in \mathcal{C}} K_c^T u_{ck} = 0 \\ -\infty, & \text{otherwise} \end{cases}$$
(6)

Thus, the Lagrange dual of the primal SOCP is:

$$\begin{aligned} & \max_{\{u_{ck}\}} \sum_{c \in \Omega^C} u_{ck}^\top f_c \\ & \text{subject to } d_k + \sum_{c \in \mathcal{C}} K_c^\top u_{ck} = 0 \end{aligned} \quad (7)$$

$$u_{ck} \in \mathcal{C}^{(m_c+1)}, \forall c \in \Omega^C$$

Expanding $u_{ck} = \begin{bmatrix} u_{c1k} \\ u_{c2k} \end{bmatrix}$, the dual problem becomes:

$$\begin{aligned} & \max_{\{u_{c1k}, u_{c2k}\}} \sum_{c \in \Omega^C} u_{c1k} (m_{ck2} - o_{ck}^\top x_k - p_{ck}^\top q_k - q_{ck}^\top \zeta_k - r_{ck}^\top \varpi_k) + u_{c2k}^\top m_{ck1} \\ & \text{subject to } \sum_{c \in \Omega^C} (n_{ck} u_{c1k} + M_{ck}^\top u_{c2k}) + d_k = 0 \\ & (u_{c1k}, u_{c2k}) \in \mathcal{C}^{(m_c+1)}, \forall c \in \Omega^C \end{aligned} \quad (8)$$

Appendix B. 公式节 (下一节)

The overall algorithm consists of three nested iterative levels: the outer AOP for upper-level decision updates, the middle ATC for coupling coordination, and the inner PC&CG for lower-level operational optimization. The total computational complexity can be expressed as a product of the iteration numbers of each level and the cost per iteration of the innermost solver.

At each AOP outer iteration, two strongly convex subproblems with decision variables $x^{(1)}$ and $x^{(2)}$ are solved alternately while fixing the other block. Each subproblem has an objective function of the form.

$$\min_z \frac{1}{2} z^\top (Q + \rho I) z + b^\top z \quad (1)$$

where ρI is the augmented penalty matrix ensuring positive definiteness. Let $r_k = \|z^{k+1} - z^k\|$ denote the residual and ε the convergence tolerance. Given the spectral bounds $\mu = \lambda_{\min}(Q + \rho I)$, $L = \lambda_{\max}(Q + \rho I)$, the convergence factor is

$$\rho_{\text{AOP}} = 1 - \mu/2L \quad (2)$$

and the required number of AOP iterations to reach convergence is [51].

$$K_{\text{AOP}} \leq \frac{\log(\varepsilon/r_0)}{\log \rho_{\text{AOP}}} \quad (3)$$

Within each AOP iteration, the ATC layer enforces consistency between the upper and lower decision variables through an augmented Lagrangian.

$$\mathcal{L}_\mu(x, y) = f(x, y) + \frac{\mu}{2} \|Ax - By\|_2^2 \quad (4)$$

where the penalty parameter updates geometrically, $\mu^{k+1} = \beta \mu^k$.

Let $g^k = \|Ax^k - By^k\|$ and stopping tolerance ε_g ; the mismatch decreases as

$$g^k \leq \beta^{-k} g^0 \quad (5)$$

Given the contraction factor $0 \leq \beta \leq 1$, the number of ATC iterations required to reduce the mismatch from g^0 to the tolerance ε_g is bounded by

$$K_{\text{ATC}} \leq \frac{\log(\varepsilon_g/g^0)}{\log \beta} \quad (6)$$

For each ATC subproblem, the embedded PC&CG procedure solves a lower-level linear convex operational model. Given n_ξ continuous variables and m_ξ active linear constraints, the feasible bases correspond to extreme points of the inner feasible region. The number of iterations before PC&CG terminates is bounded by $\binom{n_\xi + m_\xi}{m_\xi}$ [47], where $\binom{n_\xi + m_\xi}{m_\xi}$ represents the combinatorial number and

$$\binom{n_\xi + m_\xi}{m_\xi} = \frac{(n_\xi + m_\xi)!}{n_\xi! m_\xi!} \quad (7)$$

Combining the three layers, the total theoretical computational complexity is obtained as $\mathcal{O}\left[K_{\text{AOP}} \times K_{\text{ATC}} \times \binom{n_\xi + m_\xi}{m_\xi}\right]$.

Data availability

Data will be made available on request.

References

- [1] Liu X. Research on collaborative scheduling of internet data center and regional integrated energy system based on electricity-heat-water coupling. *Energy* 2024; 292:130462.

- [2] Li J, Bao Z, Li Z. Modeling demand response capability by internet data centers processing batch computing jobs. *IEEE Trans Smart Grid* 2014;6(2):737–47.
- [3] Guo C, Luo F, Cai Z, et al. Integrated energy systems of data centers and smart grids: state-of-the-art and future opportunities. *Appl Energy* 2021;301:117474.
- [4] Vafamehr A, Khodayar ME, Manshadi SD, et al. A framework for expansion planning of data centers in electricity and data networks under uncertainty. *IEEE Trans Smart Grid* 2017;10(1):305–16.
- [5] Yin X, Ye C, Ding Y, et al. Combined heat and power dispatch against cold waves utilizing responsive internet data centers. *IEEE Trans Sustain Energy* 2023;15(2): 819–34.
- [6] Liang C, Zeng B, Lei Y, et al. Market-government dual-driven framework for reliability-centric planning of a computation-energy integrated system with data centers. *Appl Energy* 2025;396:126272.
- [7] Islam MA, Ren S, Quan G, et al. Water-constrained geographic load balancing in data centers. *IEEE Trans Cloud Comput* 2015;5(2):208–20.
- [8] Chen S, Zhang Q, Zhai J, et al. MILP optimization of the multi-heat pump waste heat recovery system integrated with full-free cooling data center through lake water. *Energy* 2025;318:134823.
- [9] Su C, Wang L, Sui Q, et al. Optimal scheduling of a cascade hydro-thermal-wind power system integrating data centers and considering the spatiotemporal asynchronous transfer of energy resources. *Appl Energy* 2025;377:124360.
- [10] Oh H, Jin W, Peng P, et al. Techno-economic performance of reservoir thermal energy storage for data center cooling system. *Appl Energy* 2025;391:125858.
- [11] Islam MA, Ahmed K, Xu H, et al. Exploiting spatio-temporal diversity for water saving in geo-distributed data centers. *IEEE Trans Cloud Comput* 2016;6(3): 734–46.
- [12] Shi Q, Yan Y, Liu W, et al. Resilience assessment of urban power network considering multi-stage electrical-water fault propagation. *IEEE Trans Power Syst* 2024;40(4):2819–31.
- [13] Biswas RS, Pal A, Werho T, et al. A graph theoretic approach to power system vulnerability identification. *IEEE Trans Power Syst* 2020;36(2):923–35.
- [14] Qi S, Milojicic D, Bash C, et al. MOSAIC: A multi-objective optimization framework for sustainable datacenter management. In: *Proceedings of the 2023 IEEE 30th International Conference on High Performance Computing, Data, and Analytics (HiPC)*. 2023. p. 51–60.
- [15] Guan A, Zhou S, Gu W, et al. Cross-system fault propagation mechanism in heat-electrical integrated energy systems: insights from pipe leakage. *IEEE Trans Power Syst* 2024;39(6):7216–26.
- [16] Li B, Wang S, Chen Y, et al. Joint network expansion planning with internet data center and charging stations for EV sharing system. *IEEE Trans Ind Inform* 2024;20 (11):13251–61.
- [17] Iqbal W, Berral JL, Erradi A, et al. Real-time data center's telemetry reduction and reconstruction using Markov chain models. *IEEE Syst J* 2019;13(4):4039–50.
- [18] Zhan L, Li G, Xu C, et al. Experience knowledge decomposition–data generation: enhanced multi-step short-term cooling load predictions in data centres with data shortage issues. *Energy* 2025;328:136476.
- [19] Cai S, Xie Y, Wu Q, et al. Distributionally robust microgrid formation approach for service restoration under random contingency. *IEEE Trans Smart Grid* 2021;12(6): 4926–37.
- [20] Khodayar M, Wang J, Manthouri M. Interval deep generative neural network for wind speed forecasting. *IEEE Trans Smart Grid* 2018;10(4):3974–89.
- [21] Zeng B, Luo Y. Potential of harnessing operational flexibility from public transport hubs to improve reliability and economic performance of urban multi-energy systems: a holistic assessment framework. *Appl Energy* 2022;322:119488.
- [22] Abdul AT, Saravanas AD, Theodorou EA. Scaling robust optimization for multi-agent robotic systems: a distributed perspective. *arXiv* 2024;arXiv:2402.16227.
- [23] Giannelos S, Konstantelos I, Strbac G. Option value of demand-side response schemes under decision-dependent uncertainty. *IEEE Trans Power Syst* 2018;33: 5103–13.
- [24] Zhan Y, Zheng QP, Wang J, et al. Generation expansion planning with large amounts of wind power via decision-dependent stochastic programming. *IEEE Trans Power Syst* 2016;32:3015–26.
- [25] Su Y, Liu F, Wang Z, et al. Multi-stage robust dispatch considering demand response under decision-dependent uncertainty. *IEEE Trans Smart Grid* 2022;14: 2786–97.
- [26] Dong S, Zang T, Zhou B, Luo H, Zhou Y, Xiao X. Robust coordinated resilience enhancement strategy for communication networks of power and thermal cyber-physical systems considering decision-dependent uncertainty. *Appl Energy* 2024; 368:123494.
- [27] Zeng B, Feng J, Liu N, et al. Co-optimized parking lot placement and incentive design for promoting PEV integration considering decision-dependent uncertainties. *IEEE Trans Ind Inform* 2020;17:1863–72.
- [28] Haghighat H, Wang W, Zeng B. Robust microgrid capacity investment with endogenous and exogenous uncertainties. *IEEE Trans Smart Grid* 2023;15: 2480–92.
- [29] Zhang G, Zhang S, Zhang W, et al. Distributed energy management for multiple data centers with renewable resources and energy storages. *IEEE Trans Cloud Comput* 2020;10:2469–80.
- [30] Sharma S, Li Q. Decentralized optimization of energy-water nexus based on a mixed-integer boundary compatible algorithm. *Appl Energy* 2024;359:122588.
- [31] Wang S, Sun B, Tan X, et al. Real-time coordination of transmission and distribution networks via Nash bargaining solution. *IEEE Trans Sustain Energy* 2021;12:2238–54.
- [32] Pan C, Shao C, Hu B, et al. Modeling the reserve capacity of wind power and the inherent decision-dependent uncertainty in the power system economic dispatch. *IEEE Trans Power Syst* 2022;38:4404–17.
- [33] Hu B, Pan C, Shao C, et al. Decision-dependent uncertainty modeling in power system operational reliability evaluations. *IEEE Trans Power Syst* 2021;36: 5708–21.
- [34] Szpak D, Tchórzewska-Cieślak B, Stręk M. A new method of obtaining water from water storage tanks in a crisis situation using renewable energy. *Energies* 2024;17: 874.
- [35] Pan G, Gu W, Zhou S, et al. Synchronously decentralized adaptive robust planning method for multi-stakeholder integrated energy systems. *IEEE Trans Sustain Energy* 2019;11:1128–39.
- [36] Wang C, Gao N, Wang J, et al. Robust operation of a water-energy nexus: a multi-energy perspective. *IEEE Trans Sustain Energy* 2020;11:2698–712.
- [37] Micheleni N, Park H, Papalambros PY. Convergence properties of analytical target cascading. *AIAA J* 2003;41:897–905.
- [38] Boyd SP, Vandenberghe L. *Convex optimization*. Cambridge: Cambridge University Press; 2004.
- [39] Ma D, Cao X, Zeng B, et al. Proactive robust hardening of resilient power distribution network: decision-dependent uncertainty modeling and fast solution strategy. *IEEE Trans Autom Sci Eng* 2025;22:14953–66.
- [40] Zeng B, Wang W. Two-stage robust optimization with decision dependent uncertainty. Preprint arXiv:2203.16484 2022.
- [41] Xie M, Ji X, Hu X, et al. Autonomous optimized economic dispatch of active distribution system with multi-microgrids. *Energy* 2018;153:479–89.
- [42] Li Z, Shahidehpour M, Wu W, et al. Decentralized multiarea robust generation unit and tie-line scheduling under wind power uncertainty. *IEEE Trans Sustain Energy* 2015;6:1377–88.
- [43] Zeng B, Zhou Y, Xu X, Cai D. Bi-level planning approach for incorporating the demand-side flexibility of cloud data centers under electricity-carbon markets. *Appl Energy* 2024;357:122406.
- [44] Xiong Z, Shen X, Sun H. Two-stage robust planning for park-level integrated energy system considering uncertain equipment contingency. *IEEE Trans Smart Grid* 2025; 16(2):1112–26 [in press].
- [45] Ceseña EAM, Zhou Y, Gutierrez JNM, et al. Quantifying the impacts of modelling assumptions on accuracy and computational efficiency for integrated water-energy system simulations under uncertain climate. *IEEE Trans Sustain Energy* 2022;13: 1370–82.
- [46] Xiao Z, He D, Du J. A Stackelberg game pricing through balancing trilateral profits in big data market. *IEEE Internet Things J* 2020;8:12658–68.
- [47] Sun X, Cao X, Zeng B, et al. Stochastic-robust planning of networked hydrogen-electrical microgrids: a study on induced refueling demand. *IEEE Trans Smart Grid* 2025;16:115–30.
- [48] Alhazmi M, Dehghanian P, Nazemi M, et al. Uncertainty-informed operation coordination in a water-energy nexus. *IEEE Trans Ind Inform* 2022;19:6439–49.
- [49] Sun Y, Ding Z, Yan Y, et al. Privacy-preserving energy sharing among cloud service providers via collaborative job scheduling. *IEEE Trans Smart Grid* 2024;16(2): 1168–80.
- [50] Luo Y, Zeng B, Zhang W, et al. Coordinative planning of public transport electrification, RESs and energy networks for decarbonization of urban multi-energy systems: a government-market dual-driven framework. *IEEE Trans Sustain Energy* 2023;15:538–55.
- [51] Tseng P. Convergence of a block coordinate descent method for nondifferentiable minimization. *J Optim Theory Appl* 2001;109:475–94.